
COUPLED MULTISCALE PALEOCLIMATE RECONSTRUCTION WITH FOUR-DIMENSIONAL VARIATIONAL DATA ASSIMILATION

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ABSTRACT

Paleoclimate archives extend climate knowledge beyond the instrumental era, registering different seasons, variables, time averages, and memory lengths. A longstanding problem is to integrate these heterogeneous sources of information within a unified methodology. Here we present a new data-assimilation framework, Last Millennium Reanalysis 4D-Var (LMR4D-Var), which reconstructs climate trajectories from these heterogeneous datasets while balancing errors in the model, observations, and initial conditions. We compare results using LMR4D-Var to assimilate proxies from PAGES2k, Temp12k, and borehole temperature profiles without treating them as instantaneous equivalents. Instrumental verification shows that LMR4D-Var achieves the highest skill compared with previous reconstructions. Borehole assimilation preserves skill against withheld annually resolved records, increases agreement between reconstructed 300–2000-m ocean heat content (OHC) and independent estimates, and yields a cooler reconstructed Little Ice Age ocean. Modern 130-year trends in both reconstructed OHC layers significantly exceed their pre-1870 Common Era trend distributions. Results for Temp12k demonstrate assimilation of decadal-to-millennial records and the potential for Holocene and deeper-time applications with suitable emulators.

Plain Language Summary

Evidence of past climate comes from natural archives that record different aspects of climate over very different periods. Tree rings record individual seasons or years, while sediments can record averages over decades to centuries. Temperatures measured in boreholes retain a memory of past surface temperatures. Methods that reconstruct climate by moving forward through time without revisiting earlier estimates cannot use later observations to revise those estimates. This limits their ability to use data, such as borehole temperature profiles, that simultaneously contain information about present conditions and preceding decades or centuries. We developed LMR4D-Var to reconstruct an entire climate history jointly, combining observations with a simplified climate model while accounting for uncertainty in both. The method combines tree-ring, sediment, and borehole records to reconstruct seasonal changes in the atmosphere, ocean, and sea ice. Adding boreholes improves estimates of slowly changing deeper-ocean heat storage without reducing agreement with records that resolve individual years. In our reconstruction, ocean warming from 1870 to 2000 was faster than during any equally long interval between 500 BCE and 1869 CE. Experiments with older records suggest that this approach could reconstruct climate over the past 12,000 years with models suited to those conditions.

1 Introduction

Instrumental observations of recent climate change in the atmosphere and the ocean are too short to define the full range of natural variability and distinguish it from forced climate variability (IPCC, 2023). Paleoclimate archives provide indirect climate information, but different types of archives record different aspects of the climate system, and are

relevant to different timescales. For example, tree rings and corals (PAGES2k Consortium and others, 2017) resolve annual or seasonal temperature and precipitation, many sediment and pollen records (Kaufman et al., 2020) resolve temperature changes averaged over decades or longer, and borehole temperature profiles contain a diffused memory of past surface temperature changes over a wide range of time scales as a function of depth (Bradley, 1999; Pollack and Huang, 2000; Council et al., 2007; Jones et al., 2009; Christiansen and Ljungqvist, 2017; IPCC, 2023). A central challenge in climate reconstruction is to combine this information that is heterogeneous in space, time, and climate variable, while accounting for errors in all sources (Tierney et al., 2025; Zhang et al., 2025). A reconstruction framework that treats all records as instantaneous risks either damping the low-frequency climate signal or, conversely, distorting the higher-frequencies.

Recently, paleoclimate data assimilation has emerged as a promising framework for climate reconstruction (Hakim et al., 2016; Tierney et al., 2025; Ning et al., 2026). Most paleoclimate data-assimilation reconstructions use time-local or sequential filtering updates (Hakim et al., 2016; Tardif et al., 2019; Franke et al., 2017; Tierney et al., 2020; Osman et al., 2021; Valler et al., 2022; Erb et al., 2022; Valler et al., 2024). “Offline” approaches estimate each analysis time from a static ensemble or covariance and therefore lack an evolving dynamical memory (Okazaki et al., 2021; Emile-Geay et al., 2025), whereas “online” approaches propagate the analyzed state forward with a climate model or emulator (Perkins and Hakim, 2017, 2021; Meng and Hakim, 2024; Meng et al., 2025; Meng and Polvani, 2026). Filtering is appropriate for real-time prediction, when future observations are unavailable, but it is unnecessarily one-sided for retrospective reconstruction, for which the full paleoclimate archive is already known. Once a filter has passed an analysis time, observations dated later cannot revise that earlier state. This limitation is particularly important for temporally nonlocal archives: for example, borehole profiles (Huang et al., 2000) encode a diffusive memory of centuries of surface-temperature history, while sediment and pollen records commonly constrain multidecadal to centennial averages (Kaufman et al., 2020). Treating such records as updates at a single analysis time discards part of their temporal information and makes it difficult to combine their low-frequency constraints with the short-timescale information carried by annually resolved records.

This limitation is especially consequential for slowly responding components of the climate system. Global mean temperature (GMT) is already difficult to reconstruct prior to the instrumental era because paleoclimate observations (“proxies”) are unevenly distributed in space and time, physically selective, and are sparse, especially in many oceanic and high-latitude regions (Jones et al., 2009; PAGES2k Consortium and others, 2017). Ocean heat content (OHC) provides an even greater challenge because it integrates energy uptake over decades to centuries while ocean circulation redistributes heat. Modern ocean reconstructions and energy-budget estimates constrain recent changes (Zanna et al., 2019; Ishii et al., 2017; Wu et al., 2025), but the instrumental interval is too short to establish the full Common Era distribution of ocean heat storage, and multicentennial estimates remain sensitive to assumptions about boundary forcing and ocean circulation (Gebbie and Huybers, 2019). A central unresolved question is therefore whether the modern rate of ocean heat accumulation is distinguishable from OHC changes over comparable intervals before the instrumental era. Addressing this question requires a layer-resolved reconstruction long enough to define historical trend distributions, and a statistical comparison that accounts for the strong temporal persistence of OHC. Moreover, most paleoclimate proxies respond primarily to surface conditions rather than to subsurface ocean temperature. Reconstructing OHC separately for the 0–300-m and 300–2000-m layers therefore provides a stringent test of whether surface-sensitive archives can constrain slow ocean heat storage through an evolving coupled climate state. Such an OHC reconstruction may be viewed as a dynamically constrained estimate rather than as directly constrained by observations.

We formulate paleoclimate reconstruction as a trajectory-smoothing problem using four-dimensional variational data assimilation (4D-Var), in which the full climate trajectory is estimated conditional on all observations within the assimilation window (Le Dimet and Talagrand, 1986; Talagrand and Courtier, 1987; Fisher et al., 2005; Evensen et al., 2022). Each observation operator acts on the portion of the trajectory represented by an archive’s seasonality, averaging interval, or physical memory, and observations later in calendar time can revise earlier states through the modeled temporal relationships. This formulation allows annually resolved records to constrain short-timescale variability while low-resolution and long-memory archives constrain slower changes within the same evolving climate trajectory, without forcing these distinct observations to behave as instantaneous equivalents. Rather than analyzing each observation time-independently, 4D-Var determines the model trajectory that best fits observations distributed across the entire assimilation window (Le Dimet and Talagrand, 1986; Talagrand and Courtier, 1987; Courtier et al., 1994; Evensen et al., 2022). “Strong-constraint” 4D-Var assumes that the model is perfect, so that the best-fit trajectory is defined by the initial condition (Lorenc, 2003). “Weak-constraint” 4D-Var, which we use here, instead estimates model-error increments together with the initial state, allowing observations to correct an imperfectly modeled trajectory (Trémolet, 2006; Evensen et al., 2022). This objective explicitly balances uncertainty in the initial condition, departures from the model trajectory, and mismatch to the observations. The associated error covariances determine the relative influence of these three terms, allowing proxy records with different error characteristics to be weighted within a common optimization.

This distinction is important for paleoclimate applications because reduced-order priors and long climate-model integrations contain drift, structural bias, and model-dependent low-frequency variability, particularly in the ocean (O'Neill et al., 2016).

We develop Last Millennium Reanalysis 4D-Var (LMR4D-Var), a coupled weak-constraint framework designed to combine archives with different temporal resolutions and memory lengths. A seasonal reduced-order emulator represents the coupled evolution of the atmosphere, ocean, and sea ice, while the initial state and a sequence of model-error increments are optimized over the full assimilation window. Each proxy archive is connected to the candidate climate trajectory through an observation operator that represents what the archive records and over what interval. The framework combines PAGES2k records (version 2.0.0, see (PAGES2k Consortium and others, 2017)), many of which are seasonally or annually resolved, with Temp12k records (Kaufman et al., 2020) spanning decadal to millennial resolution. Our framework also includes terrestrial borehole temperature profiles that integrate surface-temperature history through thermal diffusion (Cuesta-Valero et al., 2021). In this way, each record constrains the climate trajectory according to its seasonal sensitivity, temporal averaging, or memory of past climate, rather than being treated as an instantaneous observation. The differentiable implementation also provides a practical route for adding nonlinear, time-averaged, or history-dependent proxy system models (Evans et al., 2013; Dolman and Laepple, 2018) without the need for traditional adjoint models specific to each archive (Widmann et al., 2010).

We evaluate LMR4D-Var using three complementary strategies: a controlled pseudo-proxy experiment, verification against withheld observations, and comparisons with external estimates of GMT and layer-resolved OHC. The pseudo-proxy experiment tests whether sparse, surface-sensitive observations can recover a known but sparsely sampled coupled climate trajectory, including OHC, that is not directly assimilated. Experiments with real proxies over the instrumental era used to evaluate whether the reconstructed GMT and layer-resolved OHC are consistent with independent observational and reconstruction products. We further determine, using a bootstrap method, whether modern 130-year OHC trends in both layers exceed the reconstructed pre-1870 Common Era trend distributions. The withholding experiments test whether low-frequency archives add information without compromising skill against annually resolved records. Finally, Temp12k-only sensitivity experiments examine how the balance between observational constraints and an imperfect reduced-order prior changes over Holocene-length windows when the climate emulator is known to have missing physics. Together, this evaluation strategy distinguishes validation of the assimilation framework from evaluation of the resulting climate reconstruction and identifies where indirect OHC constraints and long-window applications require particular caution.

2 Materials and Methods

2.1 Linear Inverse Model

Long reconstructions require a simplified dynamical model or emulator that is inexpensive enough to optimize over very long windows, but still rich enough to retain the coupled covariance structure linking the atmosphere, upper ocean, and sea ice. We choose to use a linear inverse model (LIM) as the reduced-order emulator of climate evolution (Fig. 1a). LIMs are estimated directly from lagged covariance statistics of a training ensemble and provide a compact stochastic approximation to coupled climate dynamics (Penland and Magorian, 1993a; Penland and Matrosova, 1994; Perkins and Hakim, 2020). In the present application, the LIM is not intended to replace a full climate model; rather, it provides the dynamically consistent backbone on which the weak-constraint reconstruction is built.

The reconstructed climate trajectory is represented in a reduced coupled state space derived from long climate-model integrations. Large-scale surface, sea-ice, and upper-ocean variables are projected onto a truncated empirical orthogonal function (EOF) basis using the “Statistic Analysis for Climate data in Python” (SACPYPY) package (Meng et al., 2023), and the time-evolving state is represented by the corresponding principal-component amplitudes. The reduced state includes PCs for 2-meter air temperature, sea-surface temperature, ocean heat content in the upper 300 m (OHC(0–300 m)), ocean heat content over 300–2000 m (OHC(300–2000 m)), and Northern Hemisphere sea-ice concentration and thickness. This reduced-order representation is designed to retain the dominant covariance structure of the coupled system while keeping the control dimension tractable over long assimilation windows. Separating OHC(0–300 m) and OHC(300–2000 m) allows the LIM to capture the different adjustment timescales of the upper and deeper ocean, which is important for reconstructing the slow evolution of ocean heat storage (Perkins and Hakim, 2020). Throughout the manuscript, upper-2000-m ocean heat content is diagnosed as $OHC_{0-2000} = OHC_{0-300} + OHC_{300-2000}$.

Previous work has shown the importance of leveraging several model priors (Parsons et al., 2021), so we train a separate LIM for each parent climate-model product. We use five parent products: the CESM Last Millennium Ensemble (LME) (Otto-Bliesner et al., 2016), together with PMIP simulations from MPI-ESM-P (PMIP3) (Giorgetta et al., 2013), MPI-ESM1.2-LR (PMIP4) (Mauritsen et al., 2019), MRI-ESM2-0 (PMIP4) (Yukimoto et al., 2019), and

CCSM4 (PMIP3) (Landrum et al., 2013). Each of these products yields its own LIM, for a total of five LIM priors. This design broadens the range of coupled dynamics available to the reconstruction and reduces reliance on any single model's low-frequency behavior. Before estimating each LIM, the corresponding parent trajectory is detrended so that unrealistically persistent drift or slow adjustment from initial-condition imbalance is not absorbed into the fitted linear dynamics. The LIM formulation, state-vector construction, detrending procedure, and training-data summary are given in SI Section 2 and Table S2.

The LIM is discretized on the same seasonal calendar used throughout the reconstruction, with four analysis steps per year corresponding to March–April–May (MAM), June–July–August (JJA), September–October–November (SON), and December–January–February (DJF). In discrete form,

$$\mathbf{x}_{k+1} = \mathbf{G}\mathbf{x}_k + \mathbf{n}_k, \quad (1)$$

where $\mathbf{x}_k$ is the reduced climate state at analysis step k , $\mathbf{G}$ is the LIM propagator, and $\mathbf{n}_k \sim \mathcal{N}(0, \mathbf{Q})$ represents additive model error. The LIM therefore supplies both the deterministic linear dynamics and the stochastic model-error structure used in the weak-constraint formulation below. In practice, we run the reconstruction against an ensemble of LIM priors estimated from the multiple training products listed above, so that the data assimilation can draw on a broader range of dynamical behavior than any single reduced-order emulator alone.

2.2 Weak-constraint 4D-Var implementation

The assimilation problem is formulated in weak-constraint four-dimensional variational form, which extends trajectory-level variational assimilation by including model-error increments in the control vector (Sasaki, 1970; Le Dimet and Talagrand, 1986; Talagrand and Courtier, 1987; Fisher et al., 2005; Trémolet, 2006; Fisher et al., 2011; Evensen et al., 2022). Both the initial state and the sequence of model-error increments are therefore treated as unknowns and optimized jointly. Using the seasonal LIM grid defined above, the reconstruction is optimized directly as a seasonal trajectory rather than annualized after the fact. As illustrated in Fig. 1b, the control vector contains the initial reduced state together with the sequence of model-error increments. Following the forcing or control-variable formulation of weak-constraint 4D-Var (Trémolet, 2006; Fisher et al., 2011), and the notation used in SI Section 3, the control vector is $\mathbf{v} = (\mathbf{x}_0^T, \mathbf{n}_0^T, \mathbf{n}_1^T, \dots, \mathbf{n}_{K-1}^T)^T$, where $\mathbf{x}_0$ is the initial reduced state and $\mathbf{n}_k$ are the additive model-error increments in Eq. 1. Each index k denotes one seasonal analysis step. The model-error increments are spatially correlated through $\mathbf{Q}$, but are assumed to be uncorrelated from one seasonal step to the next. The full trajectory follows from the recursive LIM dynamics, or equivalently from the linear transformation $\mathbf{x} = \mathbf{U}\mathbf{v}$ with

$$\mathbf{U} = \begin{pmatrix} \mathbf{I} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{G} & \mathbf{I} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{G}^2 & \mathbf{G} & \mathbf{I} & \cdots & \mathbf{0} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mathbf{G}^K & \mathbf{G}^{K-1} & \mathbf{G}^{K-2} & \cdots & \mathbf{I} \end{pmatrix}, \quad (2)$$

so that the initial state and all subsequent model-error increments determine a single dynamically consistent trajectory. The temporal-independence assumption for model-error increments, together with the within-step covariance represented by $\mathbf{Q}$, follows standard weak-constraint formulations (Trémolet, 2006; Fisher et al., 2011; Evensen et al., 2022). The control-space prior covariance is then compactly written as

$$\mathbf{B}_v = \text{diag}(\mathbf{C}(0), \mathbf{Q}, \mathbf{Q}, \dots, \mathbf{Q}), \quad (3)$$

with the stationary covariance $\mathbf{C}(0)$ in the first block and the discrete model-error covariance $\mathbf{Q}$ in the remaining blocks, so explicit construction of the dense trajectory covariance in state space is avoided.

Assuming Gaussian initial-state, model, and observation errors, the cost function takes the standard quadratic variational form (Le Dimet and Talagrand, 1986; Fisher et al., 2005; Trémolet, 2006; Evensen et al., 2022). It can be written compactly in control space and expanded directly into the initial-state, model-error, and observation terms, corresponding to the three-way balance sketched in Fig. 1b:

$$\begin{aligned} J(\mathbf{v}) &= \underbrace{\frac{1}{2} \mathbf{v}^T \mathbf{B}_v^{-1} \mathbf{v}}_{\text{prior penalty}} + \underbrace{\frac{1}{2} [\mathbf{y} - \mathcal{H}(\mathbf{U}\mathbf{v})]^T \mathbf{R}^{-1} [\mathbf{y} - \mathcal{H}(\mathbf{U}\mathbf{v})]}_{\text{proxy misfit}} \\ &= \underbrace{\frac{1}{2} \mathbf{x}_0^T \mathbf{C}(0)^{-1} \mathbf{x}_0}_{\text{initial-state penalty}} + \underbrace{\frac{1}{2} \sum_{k=0}^{K-1} \mathbf{n}_k^T \mathbf{Q}^{-1} \mathbf{n}_k}_{\text{model-error penalty}} + \underbrace{\frac{1}{2} [\mathbf{y} - \mathcal{H}(\mathbf{U}\mathbf{v})]^T \mathbf{R}^{-1} [\mathbf{y} - \mathcal{H}(\mathbf{U}\mathbf{v})]}_{\text{proxy misfit}}. \end{aligned} \quad (4)$$

Here $\mathbf{B}_y$ is the block-diagonal prior covariance in control space, $\mathbf{C}(0)$ is the stationary covariance of the reduced LIM state, $\mathbf{Q}$ is the discrete model-error covariance, $\mathcal{H}$ denotes the collection of proxy observation operators, and $\mathbf{R}$ is the observation-error covariance. In this form the three-way balance among the initial-state constraint, model-error penalty, and proxy misfit is explicit, while the optimization remains computationally tractable over long assimilation windows.

For sensitivity experiments, and in particular for the Temp12k-only experiments, we introduce a scalar model-error weight J_m that multiplies the model-error penalty ($J_m \cdot \frac{1}{2} \sum_{k=0}^{K-1} \mathbf{n}_k^T \mathbf{Q}^{-1} \mathbf{n}_k$). Changing J_m controls the relative weight assigned to the reduced-order model prior during optimization. Larger J_m penalizes model-error increments more strongly and keeps the trajectory closer to the LIM-implied evolution, whereas smaller J_m lowers the effective model weight and allows the trajectory to depart more freely from the prior. We do not interpret J_m as a directly observable model-error amplitude; rather, it is a sensitivity parameter that controls how strongly the optimized trajectory is penalized for departing from the reduced-order model.

Optimization is performed with a gradient-based quasi-Newton method, with L-BFGS (Liu and Nocedal, 1989) used as the default solver. Gradients are obtained through automatic differentiation of the reduced-order trajectory model and differentiable proxy operators using PyTorch (Paszke et al., 2019). This gradient-based implementation makes LMR4D-Var modular with respect to the observation operator: conventional short-memory proxy system models, history-dependent operators such as boreholes, and future nonlinear proxy models can be introduced through differentiable forward operators without deriving a hand-written adjoint sensitivity for each archive type. Practical diagnostics include cost-function reduction, gradient norms, solver convergence, and the time dependence of inferred model-error penalties.

2.3 Proxy observations and observation operators

The baseline observation network consists of temperature-sensitive PAGES2k records (PAGES2k Consortium and others, 2017), which provide the densest annually resolved information over the Common Era. Two additional proxy classes are used to introduce lower-frequency constraints: the Temp12k temperature compilation (Kaufman et al., 2020) and a recent compilation of terrestrial borehole temperature profiles (Cuesta-Valero et al., 2021). As summarized schematically in Fig. 1a and documented empirically in Fig. 2, these archives differ substantially in temporal footprint, temporal coverage, and the physical meaning of the observation, and they therefore require different observation operators within the same variational objective.

For annually resolved PAGES2k records, the observation operator is based on temperature-sensitive proxy system models (PSMs, (Evans et al., 2013)). These models are univariate, linear, and calibrated in the reduced EOF-truncated climate space used by the reconstruction. Terrestrial records are calibrated against GISTEMP instrumental surface air temperature (Lenssen et al., 2019), marine records against instrumental ERSSTv5 sea-surface temperature (Huang et al., 2017), Seasonality is determined objectively during calibration; full details of the PAGES2k PSMs and screening criteria are provided in SI Section 6.

Temp12k is treated differently: because the compilation already provides temperature anomalies and associated uncertainty estimates (Kaufman et al., 2020), it is assimilated directly as a low-frequency temperature target rather than through a separately calibrated PSM. Temp12k anomalies are referenced to the 0–2000 CE climatology so that they are consistent with the anomaly convention used for the Common Era reconstruction; see SI Section 7.

For terrestrial boreholes, the observation operator maps a reconstructed surface-temperature history directly to a present-day subsurface-temperature-anomaly profile through one-dimensional conductive diffusion in a semi-infinite medium (Huang et al., 2000; Cuesta-Valero et al., 2021). In this setting, the transient temperature field satisfies

$$\frac{\partial T}{\partial t} = \kappa \frac{\partial^2 T}{\partial z^2}, \quad (5)$$

where κ is thermal diffusivity and z is depth. If $T_s(t)$ denotes the reconstructed surface-temperature history, the predicted borehole anomaly profile is written schematically as $\mathbf{y}_{bh} = \mathcal{H}_{bh}[T_s(t)]$, or in discrete form as $\mathbf{y}_{bh} = \mathbf{M}\mathbf{T}_s$, where $\mathbf{T}_s$ is the discretized surface-temperature history and $\mathbf{M}$ contains the conductive kernels determined by depth, thermal diffusivity, and the temporal discretization of the surface history. Because the borehole response depends on the integrated thermal history rather than a single analysis time, this proxy class is particularly well suited to a variational treatment with explicit time dependence. The conductive timescale increases approximately as $t \sim Z^2/(4\kappa)$, so temperatures deeper than 100 m preferentially record multidecadal to centennial variability; accordingly, the borehole comparison excludes the shallower part of each profile and retains only depths between 100 and 300 m (see SI Section 4), which limits their influence to approximately AD 1300–1700. Borehole observation uncertainty is represented with an effective depth-dependent error model derived from an inverse-plus-forward ensemble consistency test. The resulting residual variance absorbs unresolved variance associated with both site-specific environmental disturbance and the simplified conductive forward

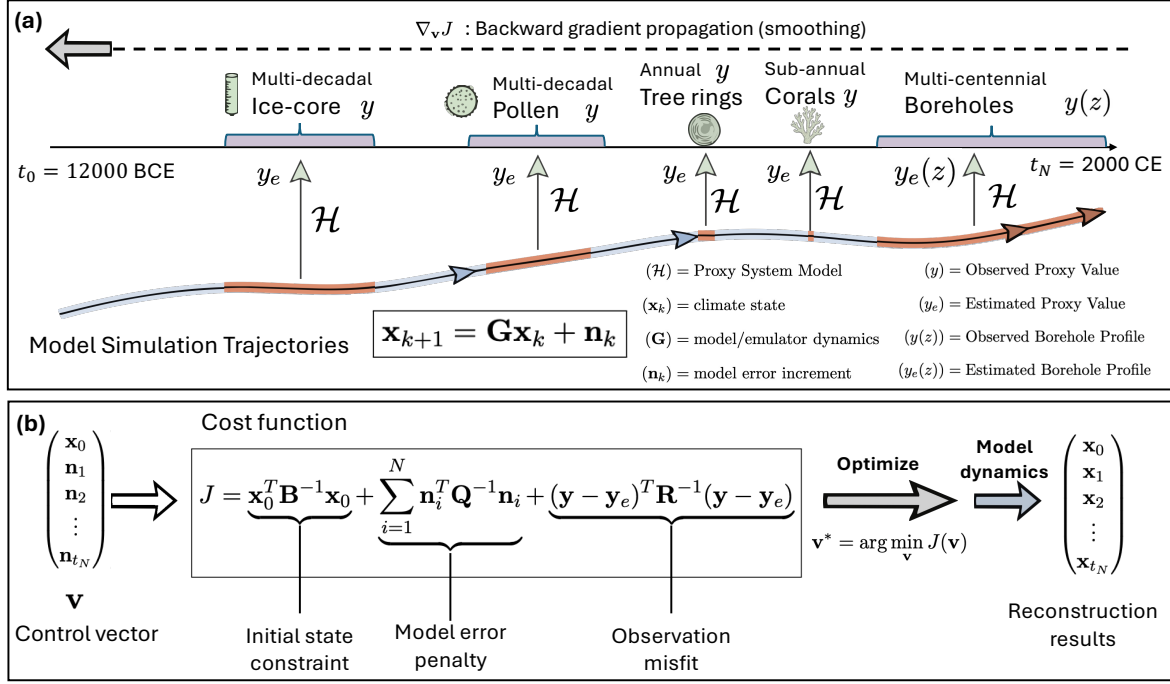


Figure 1: Multiscale weak-constraint 4D-Var framework for reconstructing past climate trajectories. (a) The reconstruction is carried out over a long assimilation window extending from 12,000 BCE to 2000 CE. Within this window, the reduced-order climate model emulator propagates the climate trajectory forward according to $\mathbf{x}_{k+1} = \mathbf{G}\mathbf{x}_k + \mathbf{n}_k$, while archive-specific observation operators $\mathcal{H}$ map the evolving climate state to observations with different temporal resolutions and memory structures, including sub-annual coral records, annual tree-ring records, multidecadal ice-core and pollen records, and multicentennial borehole profiles. The right-pointing arrows along the trajectory denote forward model propagation, whereas the dashed left-pointing arrow denotes backward gradient propagation during optimization. Because the full trajectory is optimized at once, observations later in the assimilation window can update earlier climate states and model-error increments, allowing heterogeneous observations to inform the same smoothed, dynamically constrained trajectory. (b) The control vector $\mathbf{v}$ consists of the initial condition ($\mathbf{x}_0$) and the sequence of model-error increments ($\mathbf{n}_k$). The optimized control vector is obtained by minimizing a cost function that balances three terms: the initial-condition constraint, the model-error penalty, and the observation misfit. Propagating the optimized control vector through the emulator dynamics yields the reconstructed climate trajectory.

model, and defines the diagonal borehole error covariance. The borehole observation-error covariance matrix $\mathbf{R}$ is constructed as diagonal, as in previous paleoclimate data-assimilation studies (Hakim et al., 2016; Meng et al., 2025), because the off-diagonal terms are difficult to estimate robustly in the absence of a large number of independent proxy records.

2.4 Experiments and validation strategy

We compare five reconstruction experiments that differ in the proxy classes admitted to the variational objective: P2k (PAGES2k only), P2k_BH (PAGES2k plus boreholes), P2k_T12k (PAGES2k plus Temp12k), P2k_BH_T12k (PAGES2k plus boreholes and Temp12k), and T12k (Temp12k only). This design is intended to isolate the roles of the two added low-frequency proxy classes. In particular, P2k_BH versus P2k isolates the influence of long-memory borehole constraints on Common Era temperature and ocean heat content, whereas P2k_T12k versus P2k isolates the effect of assimilating Temp12k as an additional low-frequency temperature target. P2k_BH_T12k tests whether all three proxy classes can be assimilated simultaneously, and T12k tests Holocene-scale Temp12k assimilation and sensitivity to the model-error weight. The proxy composition of each experiment is summarized in Table S3.

The five real-proxy experiments use the same five model-specific LIM priors, the same seasonal assimilation grid, and the same optimization strategy. Reconstructions are therefore obtained first as seasonal analyses with four time steps per year, and all annual or lower-frequency diagnostics shown in the main text are derived from these seasonal solutions after optimization. Differences among the five experiments can thus be attributed primarily to the observational

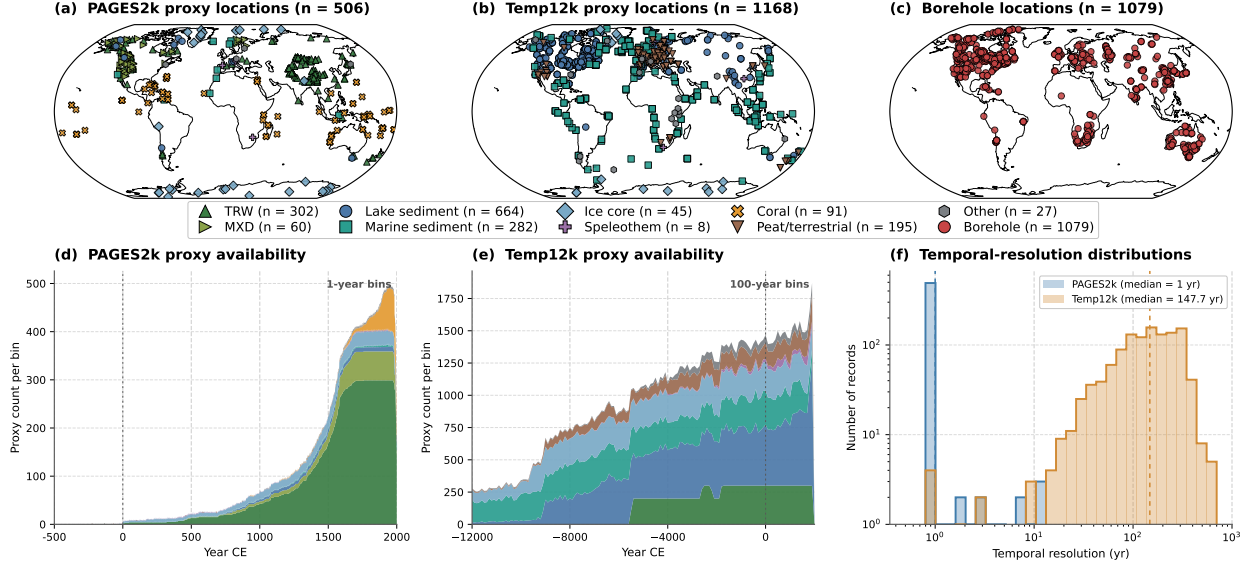


Figure 2: Spatial distribution, temporal availability, and resolution of the assimilated paleoclimate records. (a to c) Locations of the PAGES2k records (PAGES2k Consortium and others, 2017) ($n = 506$), Temp12k records (Kaufman et al., 2020) ($n = 1168$), and terrestrial boreholes (Cuesta-Valero et al., 2021) ($n = 1079$), respectively. Record types in the PAGES2k and Temp12k networks include tree-ring width (TRW; $n = 302$), maximum latewood density (MXD; $n = 60$), lake sediment ($n = 664$), marine sediment ($n = 282$), ice core ($n = 45$), speleothem ($n = 8$), coral ($n = 91$), peat and other terrestrial archives ($n = 195$), and other archives ($n = 27$); borehole locations are counted separately. (d and e) Temporal availability of PAGES2k and Temp12k records, shown in 1-year and 100-year bins, respectively, with colors indicating archive type. (f) Distributions of nominal temporal resolution for PAGES2k and Temp12k records. Dashed vertical lines indicate the median resolutions of 1 year for PAGES2k and 147.7 years for Temp12k. Together, the networks provide annually resolved, temporally averaged, and long-memory observations that constrain complementary components of climate variability across seasonal to millennial timescales.

constraints rather than to changes in model structure or solver configuration. The pseudo-proxy experiment described below is the exception: it uses four priors because the entire CESM-LME model family is excluded.

Pseudo-proxy experiment. The pseudo-proxy experiment provides a controlled test in which the true climate trajectory is known. We use CESM-LME (Otto-Bliesner et al., 2016) member 001 as the target trajectory and generate synthetic proxy observations at the same locations, seasons, and temporal resolutions as the real proxy network. For each proxy record, the corresponding observation operator is applied to the target simulation to produce a noiseless pseudo-proxy value: annually or seasonally resolved proxy system models sample the target near-surface temperature field at the calibrated season and location, Temp12k records use the same temporal averaging implied by their archive resolution, and borehole pseudo-profiles are generated by passing the target surface-temperature history through the conductive borehole forward model. Observation error is then added in observation space by drawing independent Gaussian perturbations with variance given by the corresponding diagonal element of the observation-error covariance $\mathbf{R}$; for boreholes, the perturbations use the depth-dependent borehole error variances described above. We adopt a leave-one-model-family-out design: all CESM-LME simulations are excluded from every stage of prior construction, including estimation of the EOF basis, lagged covariances, LIM propagator $\mathbf{G}$, and model-error covariance $\mathbf{Q}$. The target trajectory is reconstructed using the four remaining model-specific LIM priors, so neither member 001 nor another member of the same model family supplies dynamical information to the reconstruction. This choice emulates the structural uncertainty inherent in using a coupled general circulation model for assimilating real-world observations (Amrhein et al., 2020). Skill is evaluated by comparing the reconstructed trajectory with the known target fields for GMT, OHC(0–300 m), and OHC(300–2000 m), none of which is directly prescribed as an assimilated OHC observation.

External GMT and OHC validation. Comparisons of reconstructed GMT and OHC with instrumental and external estimates are evaluated using correlation and coefficient of efficiency (CE) over their common intervals. Following Nash and Sutcliffe (1970), CE is defined as

$$CE = 1 - \frac{\sum_t (\hat{y}_t - y_t)^2}{\sum_t (y_t - \bar{y})^2},$$

where y_t is the validation target, $\hat{y}_t$ is the corresponding reconstruction estimate, and $\bar{y}$ is the temporal mean of the validation target over the evaluation interval. A value of $CE = 1$ indicates perfect agreement, whereas $CE < 0$ indicates less skill than using $\bar{y}$ as the prediction.

OHC trend comparison. We determine whether the reconstructed 1870–2000 CE OHC trends exceed trends over all comparable intervals before 1870 CE using the bootstrap method. The analysis uses the unfiltered annual median of the five model-emulator-specific P2k_BH_T12k reconstructions. The modern interval spans 130 years and contains 131 annual values, including both endpoints. For each OHC layer, we enumerate every complete historical window containing the same number of annual values within 500 BCE–1869 CE and calculate its linear trend, yielding 2,240 historical windows. We calculate the one-sided empirical p value as $p = (b + 1)/(N + 1)$, where b is the number of historical trends greater than or equal to the modern trend and N is the total number of historical windows. The two layer-specific p values are adjusted using the Holm procedure (Holm, 1979). These empirical p values quantify the position of the modern trend within the reconstructed historical distribution.

Independent validation. For each proxy family (PAGES2k, Temp12k, and boreholes), 20% of the available records are withheld from assimilation and reserved for validation, and the withholding experiment is repeated 20 times, as in Hakim et al. (2016). Skill for withheld PAGES2k and Temp12k records is evaluated with correlation over the relevant Common Era intervals, whereas withheld borehole profiles are evaluated with root mean square error. This setup is designed to answer the central validation question raised by the main results: whether adding boreholes and Temp12k changes the reconstruction because it adds physically meaningful low-frequency information, rather than merely increasing flexibility. In particular, the independent tests ask whether added low-frequency constraints degrade skill against annually resolved records and whether they improve validation against boreholes themselves.

3 Results

LMR4D-Var reconstructs a seasonal climate trajectory by combining proxy observations with a reduced-order coupled emulator (Fig. 1). The state vector includes surface temperature, sea-surface temperature, sea ice, and two OHC layers, 0–300 m and 300–2000 m. The reconstruction is repeated with five climate-model-specific emulators (SI Section 2 and Table S2), and we compare five proxy-assimilation experiments: P2k (PAGES2k only), P2k_BH (PAGES2k plus boreholes), P2k_T12k (PAGES2k plus Temp12k), P2k_BH_T12k (PAGES2k plus boreholes and Temp12k), and T12k (Temp12k only; Table S3). The proxy network illustrates why this design is needed: PAGES2k supplies the densest annually resolved Common Era information, Temp12k extends the temporal coverage with records whose resolution ranges from decadal to millennial, and boreholes provide spatially sparse but long-memory constraints on integrated surface-temperature history (Fig. 2). This design separates the information provided by annually resolved records, low-frequency temperature targets, and long-memory borehole profiles.

We first evaluate the coupled reconstruction system in a controlled pseudo-proxy experiment. CESM-LME (CESM Last Millennium Ensemble, (Otto-Bliesner et al., 2016)) member 001 is treated as the target climate history or “ground truth”, and proxy values are generated from that target using the same proxy locations and observation operators. To provide a leave-one-model-family-out test, the entire CESM-LME ensemble is excluded; the target is reconstructed using the four remaining model-specific priors. This experiment is not an independent validation of the real-world reconstruction, but it tests whether the framework can recover known GMT and two-layer upper-ocean heat-content variability from sparse proxy-like information with an imperfect model. The reconstructed ensemble captures the main low-frequency evolution of GMT ($r = 0.71$), OHC(0–300 m) ($r = 0.69$), and OHC(300–2000 m) ($r = 0.71$) over 1000–2000 CE (Fig. S1). This agreement is encouraging for a proxy-like network that does not directly observe OHC, and the reconstruction recovers several major low-frequency departures in the target simulation, including responses associated with volcanic forcing. We therefore use OHC as an informative but demanding coupled diagnostic, and interpret it below with external comparisons and layer-resolved uncertainty.

With this controlled test as context, the P2k_BH_T12k reconstruction provides the main Common Era application of the framework. From 500 BCE to 2000 CE, reconstructed GMT and upper-2000-m OHC both show coherent low-frequency variability, including warm conditions during the Roman Warm Period (RWP, ca. 250 BCE–400 CE) and Medieval Warm Period (MWP, ca. 950–1250 CE) and colder conditions during the Little Ice Age (LIA, ca. 1400–1700 CE) (Fig. 3). Over the instrumental era, reconstructed GMT agrees closely with Berkeley Earth (Rohde and Hausfather, 2020) ($r = 0.95$, $CE = 0.89$), and over the last four centuries it is also broadly consistent with the geothermal reconstruction of Cuesta-Valero et al. (2021), despite the different treatment of borehole information in the two approaches. After linear detrending, GMT remains consistent with Berkeley Earth ($r = 0.84$, $CE = 0.70$), showing that the agreement is not solely attributable to the common long-term warming trend.

Comparison with earlier paleoclimate data-assimilation products further contextualizes the instrumental-era GMT skill (Table 1). To reduce differences arising from the choice of climate model, rather than proxy information, we

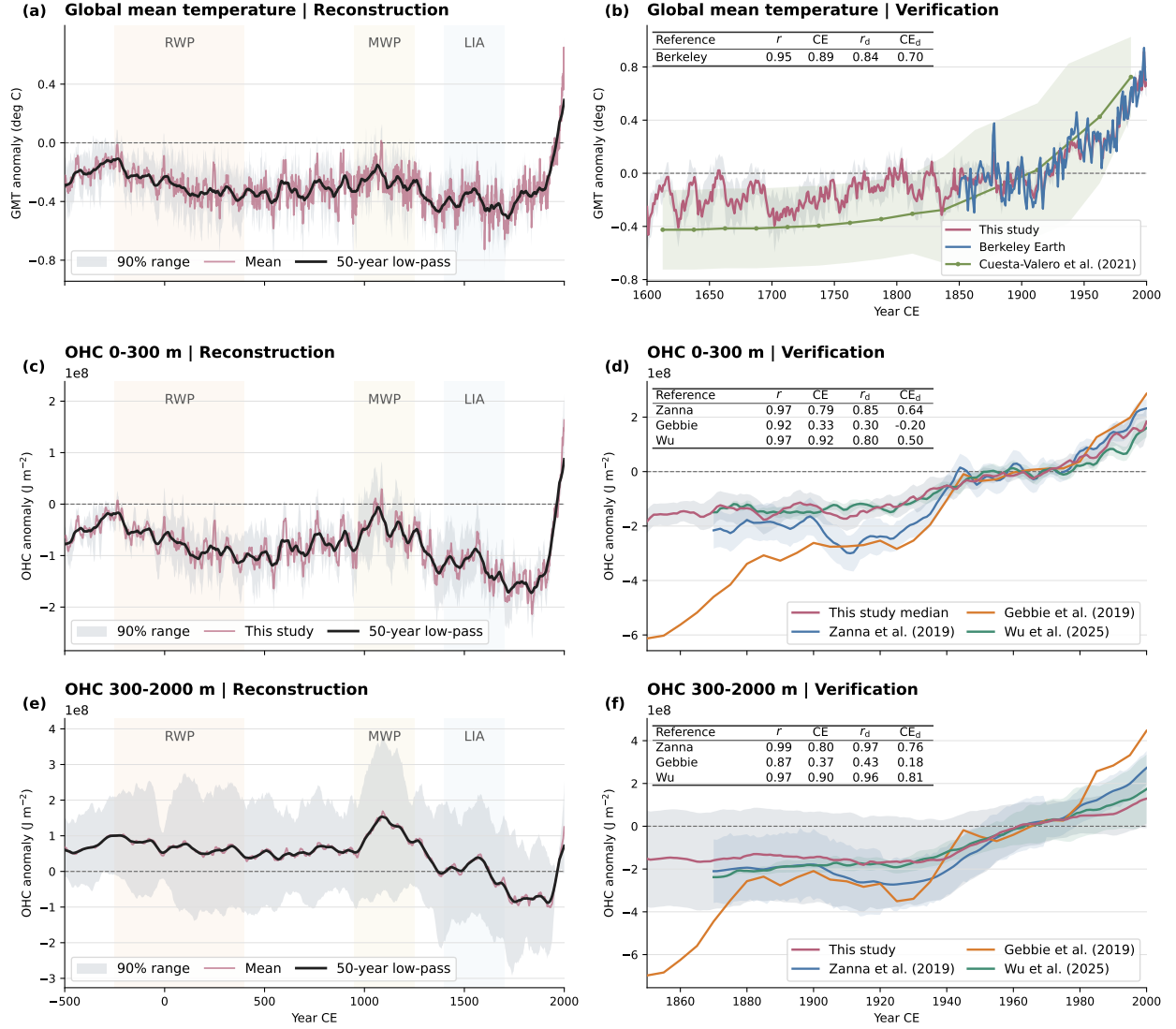


Figure 3: Reconstructions and external verification for global mean temperature and layer-resolved upper-ocean heat content. Results are from the P2k_BH_T12k experiment. (a) Reconstructed global mean temperature (GMT) anomaly from 500 BCE to 2000 CE. (b) Comparison with Berkeley Earth (Rohde and Hausfather, 2020) over the instrumental era and the geothermal reconstruction of Cuesta-Valero et al. (2021) since 1600 CE. (c and d) Reconstruction and external verification, respectively, of global ocean heat content (OHC) anomalies in the 0–300-m layer. (e and f) Corresponding reconstruction and external verification for the 300–2000-m layer. OHC estimates are compared with the external reconstructions of Zanna et al. (2019), Gebbie and Huybers (2019), and Wu et al. (2025). Muted rose curves show the mean across the five model-emulator-specific reconstructions, and gray shading denotes the nominal 90% inter-model-emulator range, defined as the mean ± 1.645 standard deviations. Black curves in the reconstruction panels show the 50-year low-pass-filtered mean. In the verification panels, Berkeley Earth and Zanna et al. (2019) are shown in blue, Cuesta-Valero et al. (2021) in olive green, Gebbie and Huybers (2019) in orange, and Wu et al. (2025) in green. Shading around the external estimates indicates reported uncertainties where available. Inset tables in (b), (d), and (f) report Pearson correlation coefficients (r), coefficients of efficiency (CE), and their detrended counterparts (r_d and CE_d). Detrended metrics are calculated after linearly detrending each reconstruction-reference pair over its common interval. Roman Warm Period (RWP; ca. 250 BCE–400 CE), Medieval Warm Period (MWP; ca. 950–1250 CE), and Little Ice Age (LIA; ca. 1400–1700 CE) intervals are marked in (a), (c), and (e).

Table 1: Instrumental-era global mean temperature skill relative to Berkeley Earth (Rohde and Hausfather, 2020). Pearson correlation coefficients (r) and coefficients of efficiency (CE) compare annual GMT anomalies from each reconstruction with Berkeley Earth over 1850–2000 CE after removing the 1950–1980 climatology. Metrics are reported for the original series (r , CE) and after separately removing a linear trend from each reconstruction and Berkeley Earth over the comparison interval (r_d , CE_d). Previously published products are listed in chronological order: PHYDA (Steiger et al., 2018), LMR v2.1 (Tardif et al., 2019), LMR Online (Perkins and Hakim, 2021), and LMR Seasonal (Meng et al., 2025), followed by the four experiments from this study. Assimilation methods include offline and online ensemble Kalman filtering (EnKF) and weak-constraint 4D-Var. The update interval is given for the online methods, and the model/emulator column identifies the climate-model information used by each reconstruction. Comparisons among the CCSM4-based rows provide the closest assessment of differences among assimilation frameworks, although the observation networks and implementation details are not identical. Within LMR4D-Var, P2k, P2k_BH, and P2k_BH_T12k use the CCSM4-trained emulator and differ in the observations assimilated; P2k_BH_T12k mean averages the five model-emulator-specific reconstructions. PHYDA uses CESM-based model information.

Series	Assimilation method	Model/emulator	r	CE	r_d	CE_d
PHYDA	Offline EnKF	CESM	0.90	0.78	0.73	0.43
LMR v2.1	Offline EnKF	CCSM4	0.93	0.80	0.80	0.63
LMR Online	Online EnKF (annual)	CCSM4	0.90	0.80	0.70	0.45
LMR Seasonal	Online EnKF (seasonal)	CCSM4	0.93	0.77	0.82	0.67
P2k	Weak-constraint 4D-Var	CCSM4	0.94	0.87	0.82	0.67
P2k_BH	Weak-constraint 4D-Var	CCSM4	0.94	0.88	0.82	0.67
P2k_BH_T12k	Weak-constraint 4D-Var	CCSM4	0.94	0.88	0.83	0.68
P2k_BH_T12k mean	Weak-constraint 4D-Var	Five models	0.95	0.89	0.84	0.70

emphasize the CCSM4 LMR4D-Var experiments because LMR v2.1, LMR Seasonal, and LMR Online also use CCSM4-based prior information (Tardif et al., 2019; Meng et al., 2025; Perkins and Hakim, 2021). Holding the parent model family in common makes these comparisons more directly interpretable. Measuring performance in terms of GMT within this CCSM4-based comparison, P2k gives $r = 0.94$ and $CE = 0.87$, while adding boreholes in P2k_BH leaves the correlation unchanged and increases CE only slightly to 0.88. P2k_BH_T12k with the CCSM4 emulator gives the same $r = 0.94$ and $CE = 0.88$. The five-model P2k_BH_T12k mean, shown separately to evaluate the effect of aggregating model-specific reconstructions, gives the highest central-estimate scores ($r = 0.95$, $CE = 0.89$). The earlier reconstructions have correlations of 0.90–0.93 and CE values of 0.77–0.80. After linear detrending, the CCSM4-based LMR4D-Var experiments retain substantial agreement with Berkeley Earth ($r_d = 0.82$ –0.83, $CE_d = 0.67$ –0.68), comparable to LMR Seasonal ($r_d = 0.82$, $CE_d = 0.67$). The five-model P2k_BH_T12k mean achieves the highest detrended scores among the products compared here ($r_d = 0.84$, $CE_d = 0.70$; Table 1). These results show that reconstruction skill extends beyond the shared linear warming trend. These central-estimate comparisons indicate that LMR4D-Var preserves, and in this evaluation modestly improves, instrumental-era GMT skill while adding long-memory observations and coupled ocean variables. The shared CCSM4 basis reduces one source of potential difference and provides a more controlled shared-prior comparison, although the products still differ in their observation networks, calibration choices, and assimilation details.

Validating OHC tests the algorithm’s ability to reconstruct an indirect field, because the assimilated proxies primarily constrain surface climate. Results show that reconstructed OHC late historical evolution is consistent in magnitude and trend with independent estimates based on historical ocean reconstruction and energy-budget methods (Zanna et al., 2019; Wu et al., 2025). For OHC(0–300 m), the comparison with Wu et al. (2025) yields $r = 0.97$ and $CE = 0.92$, and for OHC(300–2000 m) it yields $r = 0.97$ and $CE = 0.90$. To distinguish agreement in the historical trend from agreement in residual variability, we also compute r and CE after linear detrending. The detrended comparison with Wu et al. (2025) remains strong in the 300–2000-m layer ($r = 0.96$, $CE = 0.81$), but is weaker in the 0–300-m layer ($r = 0.80$, $CE = 0.50$), indicating that the common warming trend contributes more to the upper-layer scores. Detrended agreement with Zanna et al. (2019) also remains high after detrending for both layers ($r = 0.85$, $CE = 0.64$ for 0–300 m; $r = 0.97$, $CE = 0.76$ for 300–2000 m), whereas the corresponding comparison with Gebbie and Huybers (2019) is substantially weaker ($r = 0.30$, $CE = -0.20$; $r = 0.43$, $CE = 0.18$). Layer-resolved spatial comparisons with Wu et al. (2025) show stronger positive correlations over the Southern Ocean and the South Pacific, Indian, and South Atlantic sectors, but weaker correlations over the North Atlantic and Pacific (Fig. S3). These spatial correlations reflect consistent trend patterns: the 0–300-m trends agree most strongly over the Southern Ocean, Indian Ocean, and Atlantic sectors, whereas the 300–2000-m trends are most consistent over the Southern Hemisphere, especially the extratropical Southern Ocean (Fig. S4). Comparisons across the four proxy configurations show how each archive class affects agreement with Wu et al. (2025) (Fig. 4). In the 0–300-m layer, the reconstructed trajectories are nearly identical, with

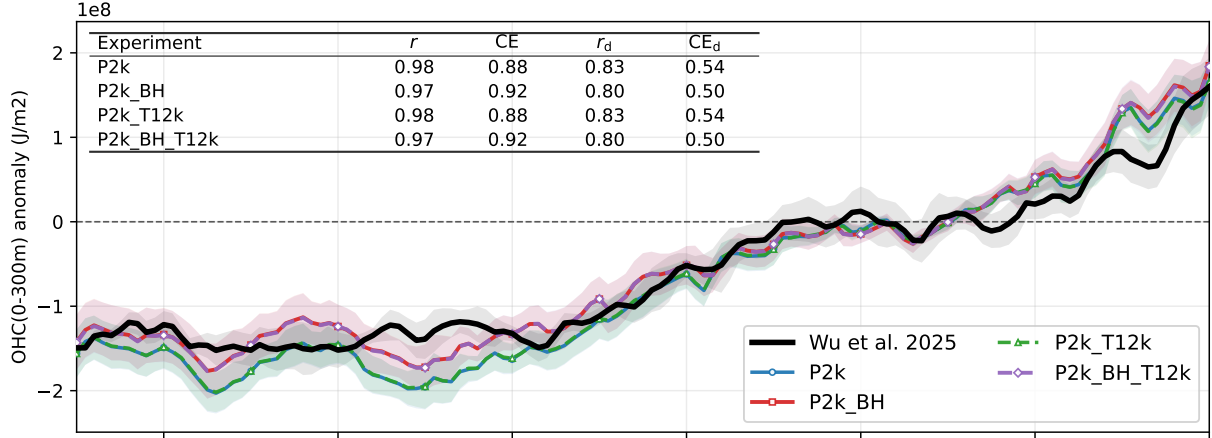
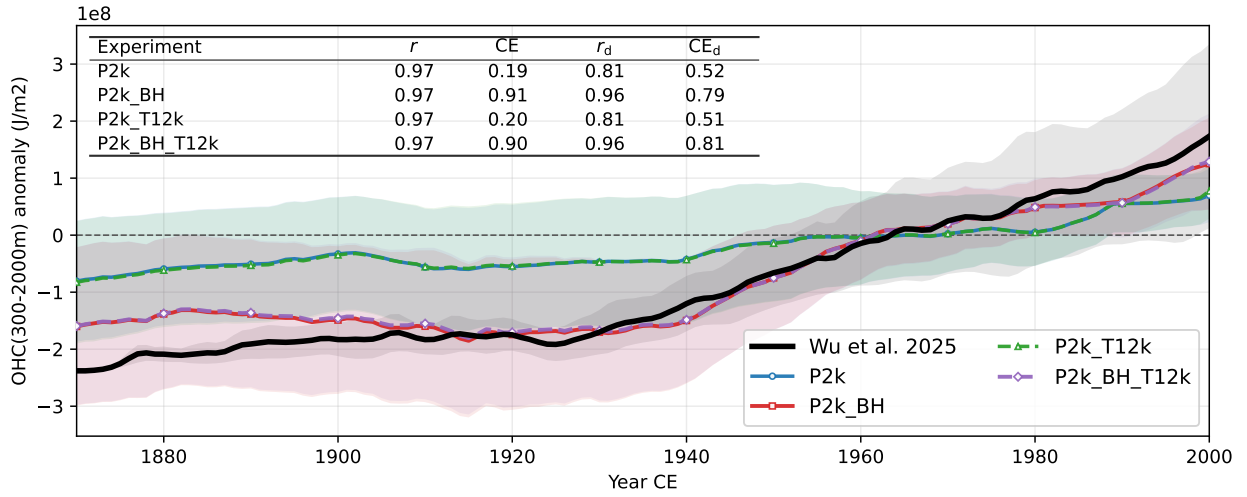
(a) Comparison of OHC(0-300 m) reconstructions with Wu et al. (2025)**(b) Comparison of OHC(300-2000 m) reconstructions with Wu et al. (2025)**

Figure 4: Borehole assimilation improves agreement with an external reconstruction of deeper-ocean heat content. Global-mean ocean heat content (OHC) anomalies from four LMR4D-Var experiments compared with the estimate of Wu et al. (2025) over 1870–2000 CE. (a) OHC anomalies for the 0–300-m layer. (b) OHC anomalies for the 300–2000-m layer. Colored lines show the mean across the five model-emulator-specific reconstructions for the P2k, P2k_BH, P2k_T12k, and P2k_BH_T12k experiments; colored shading denotes ± 1 standard deviation across these reconstructions. In panel (a) and (b), P2k closely overlaps P2k_T12k, whereas P2k_BH closely overlaps P2k_BH_T12k, producing two visually distinct pairs. Black lines show the external estimate, with gray shading indicating its reported uncertainty. All anomalies are referenced to the 1950–1980 CE climatology. Pearson correlation coefficients (r) and coefficients of efficiency (CE), and their detrended counterparts (r_d and CE_d), reported in the tables, are calculated against the same external estimate over 1870–2000 CE. Detrended metrics are calculated after linearly detrending each reconstruction-reference pair over its common interval.

CE ranging from 0.88 to 0.92 and detrended CE from 0.50 to 0.54. The borehole effect is much larger in the 300–2000-m layer: CE increases from 0.19–0.20 without boreholes to 0.90–0.91 with boreholes, while detrended CE increases from 0.51–0.52 to 0.79–0.81. Adding Temp12k produces little additional change over the instrumental interval, as P2k overlaps P2k_T12k and P2k_BH overlaps P2k_BH_T12k. The borehole-related improvement is also evident against Gebbie and Huybers (2019), with CE increasing from 0.18 to 0.60–0.61 and detrended CE from 0.25 to 0.47. Against Zanna et al. (2019), CE increases from 0.12–0.13 to 0.80–0.81 and detrended CE from 0.27–0.28 to 0.76–0.77 (Fig. S5). These comparisons associate the improved deep-ocean agreement primarily with long-memory borehole information transmitted through the coupled emulator covariance. Support is therefore strongest for large-scale OHC evolution, while regional and depth-dependent patterns remain less certain.

The reconstructed modern OHC trends significantly exceed pre-1870 Common Era trends in both the 0–300-m and the 300–2000-m layers (Fig. 5). The 1870–2000 CE trends are $2.302 \times 10^8 \text{ J m}^{-2} \text{ century}^{-1}$ for OHC(0–300 m) and $2.136 \times 10^8 \text{ J m}^{-2} \text{ century}^{-1}$ for OHC(300–2000 m). None of the 2,240 complete 130-year historical windows within 500 BCE–1869 CE has a trend as large as the modern trend in either layer. Both layers therefore have a one-sided empirical $p = 0.000446$ and a Holm-adjusted $p = 0.000892$. The modern trends are also consistent with the estimates of Zanna et al. (2019) and Wu et al. (2025). By contrast, the trends implied by Gebbie and Huybers (2019) are much larger in both layers. The layer-resolved comparison with Gebbie and Huybers (2019) (Fig. S6) highlights the remaining uncertainty in multicentennial OHC amplitude: that reconstruction implies variability that is approximately an order of magnitude larger than our estimate in both layers, likely reflecting differences in boundary forcing assumptions, circulation constraints, and the propagation of surface anomalies into subsurface heat storage. One likely contributor to the smaller amplitude in our reconstruction is the prior itself. The last-millennium model simulations used to train the climate emulators contain relatively small global-mean OHC variability, on the order of 10^8 J m^{-2} , and also contain strong long-term drifts, especially in OHC(300–2000 m) (Fig. S7). Such drifts may arise from incomplete equilibration of the initial deep-ocean state and the long adjustment timescale of the deeper ocean, consistent with the importance of deep-ocean initial conditions for low-frequency climate variability proposed by Zhu et al. (2019). Because the emulators are trained after linear detrending, the procedure removes unrealistic drift but may also remove multicentennial variability. In addition, the emulators represent coupled climate evolution using a time-invariant linear propagator estimated in a truncated EOF space. The linear dynamics may not fully capture state-dependent changes in ocean circulation, mixing, and heat uptake when strong external forcing drives the climate outside the range of variability represented in the last-millennium training simulations. Moreover, although EOF truncation retains the dominant modes of variability, it may omit lower-variance spatial or vertical structures that contribute to the forced OHC response, particularly in the deeper ocean. These approximations may attenuate the amplitude or alter the spatial and vertical distribution of reconstructed OHC, although their relative contributions have not been quantified here. Even with these limitations, the historical-period OHC trends remain close to the estimates of Wu et al. (2025), indicating that the framework retains skill for recent forced changes while leaving substantial uncertainty in multicentennial OHC variability.

The layer-resolved maps also show that OHC changes across different periods have different spatial structures. The MWP-to-LIA change in OHC(300–2000 m) has a pronounced North Pacific component (Fig. S8), whereas the modern OHC(300–2000 m) increase in Wu et al. (2025) does not show the same North Pacific-centered structure (Fig. S4). One possible explanation is that the two intervals reflect different forcing regimes. The MWP-to-LIA transition includes strong volcanic forcing and associated dynamical responses (Mann et al., 2009; Toohey and Sigl, 2017), whereas the modern increase is dominated by anthropogenic greenhouse-gas forcing (Gleckler et al., 2012; IPCC, 2021). Different forced-response patterns can project onto different modes of ocean circulation and heat redistribution, which may in turn produce distinct OHC patterns. We therefore treat this contrast as a hypothesis-generating result rather than formal attribution, because targeted forced-simulation experiments are needed to test whether the inferred pattern difference reflects volcanic forcing, greenhouse-gas forcing, internal variability, or model-dependent covariance structure.

Additionally, pairwise comparisons among the reconstruction experiments show that the different proxy classes affect the trajectory in ways that are consistent with their temporal information content (Figs. S9 and S10). Reconstructions that include PAGES2k share very similar GMT evolution, with pairwise correlations generally exceeding 0.95. Adding boreholes or Temp12k therefore does not substantially change the main Common Era surface-temperature variability. OHC comparisons are also highly correlated across P2k, P2k_BH, P2k_T12k, and P2k_BH_T12k, but show larger amplitude differences, especially when boreholes are included, consistent with OHC being more sensitive to low-frequency constraints. In contrast, the T12k reconstruction has substantially lower correlations with the PAGES2k-constrained experiments (r is around 0.5), indicating that Temp12k captures broad low-frequency structure but does not replace the higher-resolution Common Era constraints provided by PAGES2k.

The borehole experiments provide the clearest example of how long-memory archives alter the reconstruction. Relative to the P2k baseline, P2k_BH produces an additional global-mean LIA cooling of approximately 0.04 K, with a standard deviation of 0.01 K across the five model-emulator-specific reconstructions (Fig. 6). Borehole assimilation produces a more pronounced response in ocean heat storage: global-mean LIA OHC decreases by approximately $0.15 \times 10^8 \text{ J m}^{-2}$ in the 0–300-m layer and $0.20 \times 10^8 \text{ J m}^{-2}$ in the 300–2000-m layer. The corresponding standard deviations across the five model-emulator-specific reconstructions are 0.06×10^8 and $0.07 \times 10^8 \text{ J m}^{-2}$, respectively. The pronounced OHC response is consistent with the role of boreholes as long-memory constraints: because borehole profiles integrate surface-temperature history over decades to centuries, they are especially informative about slowly varying quantities such as OHC. We note that this response in the reconstruction arises naturally, without any explicit linkage between borehole proxies and OHC.

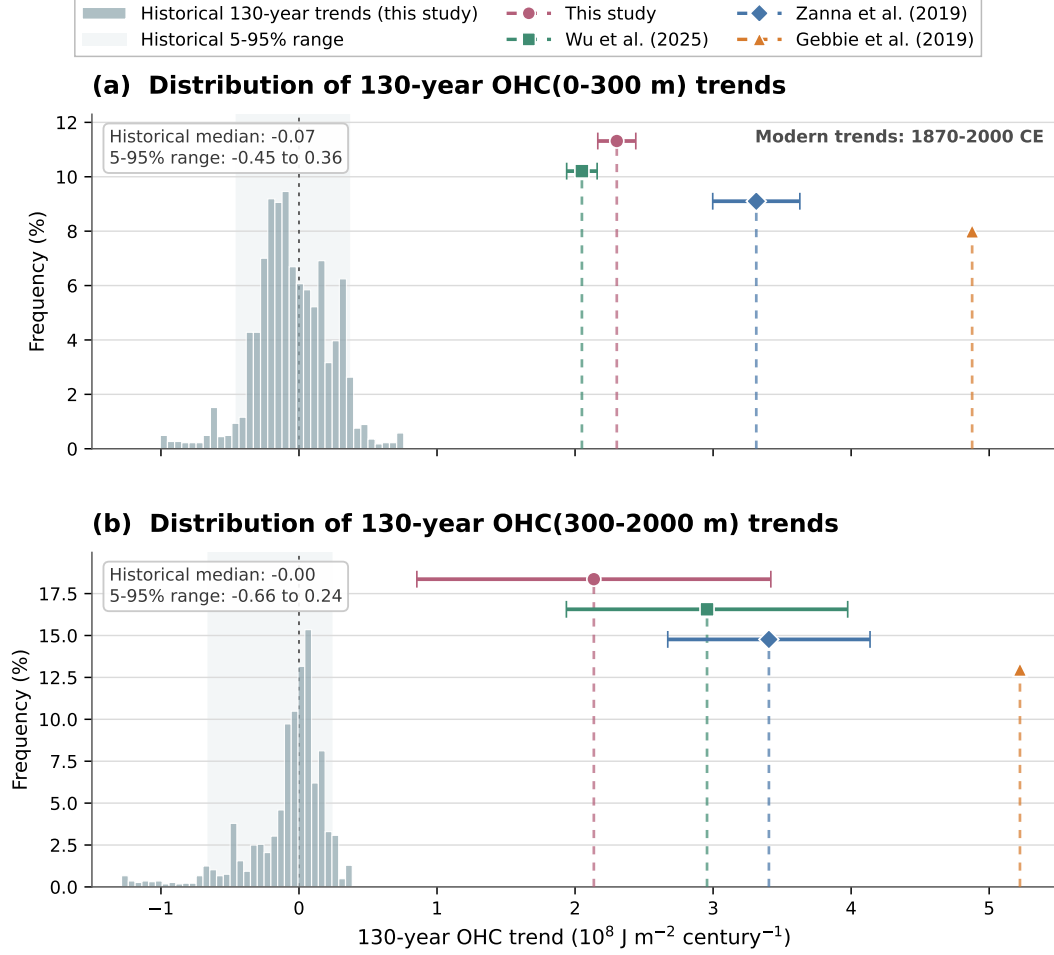


Figure 5: Modern layer-resolved ocean heat-content trends significantly exceed reconstructed pre-1870 Common Era trends. Histograms show all 130-year linear trends in reconstructed ocean heat content anomalies for (a) 0–300 m and (b) 300–2000 m, calculated from 130-year windows contained within 500 BCE–1869 CE using the annual median of the five model-emulator-specific P2k_BH_T12k reconstructions. Gray shading indicates the historical 5th–95th percentile ranges, and gray dotted lines indicate the historical medians. Colored markers indicate the 1870–2000 CE trends from this study (magenta circles), [Wu et al. \(2025\)](#) (green squares), [Zanna et al. \(2019\)](#) (blue diamonds), and [Gebbie and Huybers \(2019\)](#) (orange triangles). Dashed vertical lines connect the estimates to the horizontal axis; marker heights are used only for visual separation. Horizontal error bars denote available 90% uncertainty intervals. For this study, the nominal 90% interval is the median ± 1.645 standard deviations across the five model-emulator-specific reconstructions. Neither layer contains a historical-window trend as large as its modern trend; the one-sided empirical comparison gives Holm-adjusted $p = 0.000892$ for both layers. Trends are expressed in units of $10^8 \text{ J m}^{-2} \text{ century}^{-1}$.

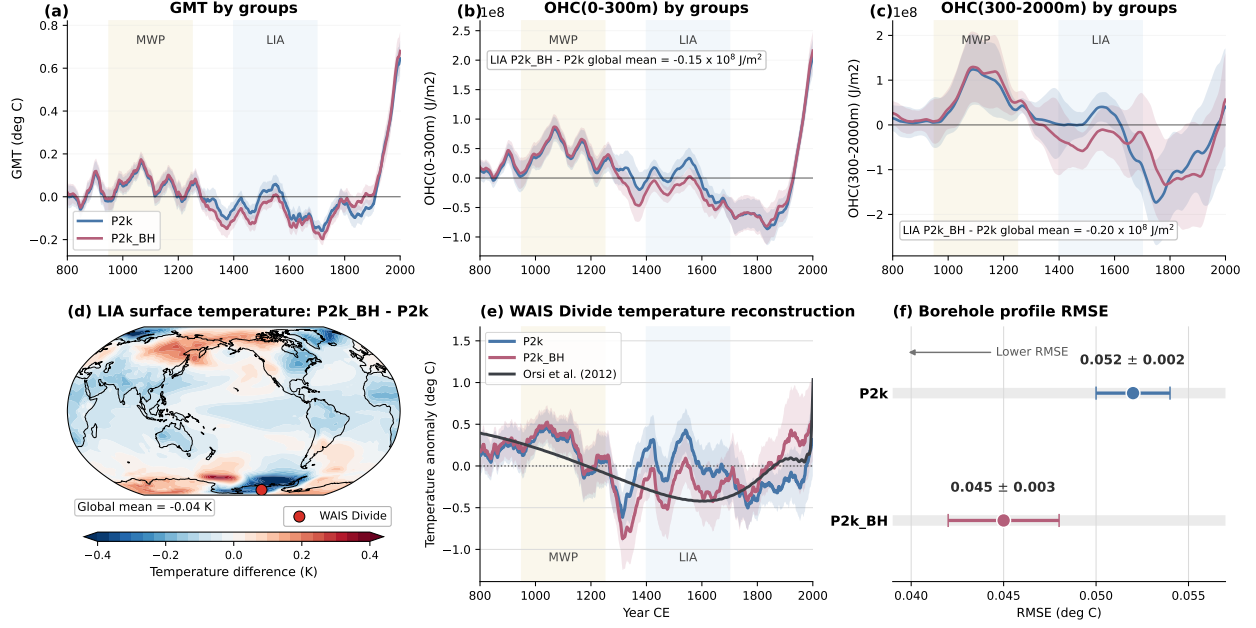


Figure 6: Borehole assimilation modifies the long-term evolution of temperature and ocean heat content and improves agreement with an independent West Antarctic borehole constraint. (a to c) Reconstructed global mean temperature (GMT), ocean heat content (OHC) in the 0–300-m layer, and OHC in the 300–2000-m layer for P2k (PAGES2k only) and P2k_BH (PAGES2k plus terrestrial boreholes), respectively. Lines show the mean across five model-emulator-specific reconstructions, and shading denotes ± 1 standard deviation across these reconstructions. Anomalies are referenced to the 0–2000 CE climatology. Values reported in (b) and (c) give the global-mean P2k_BH minus P2k differences during the Little Ice Age (LIA). (d) Spatial distribution of the LIA surface-temperature difference between P2k_BH and P2k, calculated as the 1400–1700 CE mean. The global-mean difference is reported in the panel, and the WAIS Divide site is marked by a red dot. (e) Reconstructed temperature anomalies at WAIS Divide for P2k and P2k_BH compared with the independent borehole reconstruction of Orsi et al. (2012). All experimental time series in (a) to (c) and (e) are smoothed with a 50-year low-pass filter. Shading in (e) denotes ± 1 standard deviation across the five model-emulator-specific reconstructions. (f) Root-mean-square error (RMSE) of the reconstructed borehole temperature profiles at WAIS Divide. Markers show the mean RMSE across the five model-emulator-specific reconstructions, and horizontal error bars denote ± 1 standard deviation; lower RMSE indicates closer agreement with the observed borehole profile. Shaded intervals mark the MWP (950–1250 CE) and LIA (1400–1700 CE).

The borehole influence is not purely local. The P2k_BH minus P2k LIA surface-temperature pattern has a La Niña-like structure and reaches its largest amplitude over West Antarctica. Note that we do not include the temperatures from ice in our assimilation, because doing so would require the further complexity of accounting for ice advection; we assimilate only terrestrial boreholes from Cuesta-Valero et al. (2021). An independent ice-core borehole record from West Antarctica (Orsi et al., 2012; Steig and Orsi, 2013) therefore provides an out-of-sample check. When the reconstructed temperature histories are passed through the borehole forward model (Orsi et al., 2012) at that site, the RMSE decreases from 0.052 ± 0.002 K in P2k to 0.045 ± 0.003 K in P2k_BH (Fig. 6). Although this single-site comparison does not provide a formal significance test, the reduction in RMSE provides an independent consistency check: assimilating boreholes makes the LIA temperature reconstruction more consistent with an independent long-memory constraint, supporting the interpretation that the large-scale cooling is not simply an overfit to local borehole information.

The corresponding LIA OHC differences show that the long-memory surface constraint also projects onto OHC. Borehole assimilation produces a cooler global-mean LIA ocean state in both OHC layers (Fig. 6). Because the detailed regional OHC anomalies are inferred indirectly through emulator-derived cross-variable covariance and lack independent validation, we focus on the sign and magnitude of the global-mean response rather than interpreting the spatial pattern as a robust dynamical fingerprint. Together, these results help reconcile earlier differences between borehole-based reconstructions, which tend to imply greater LIA cooling than reconstructions based primarily on annually resolved natural proxies.

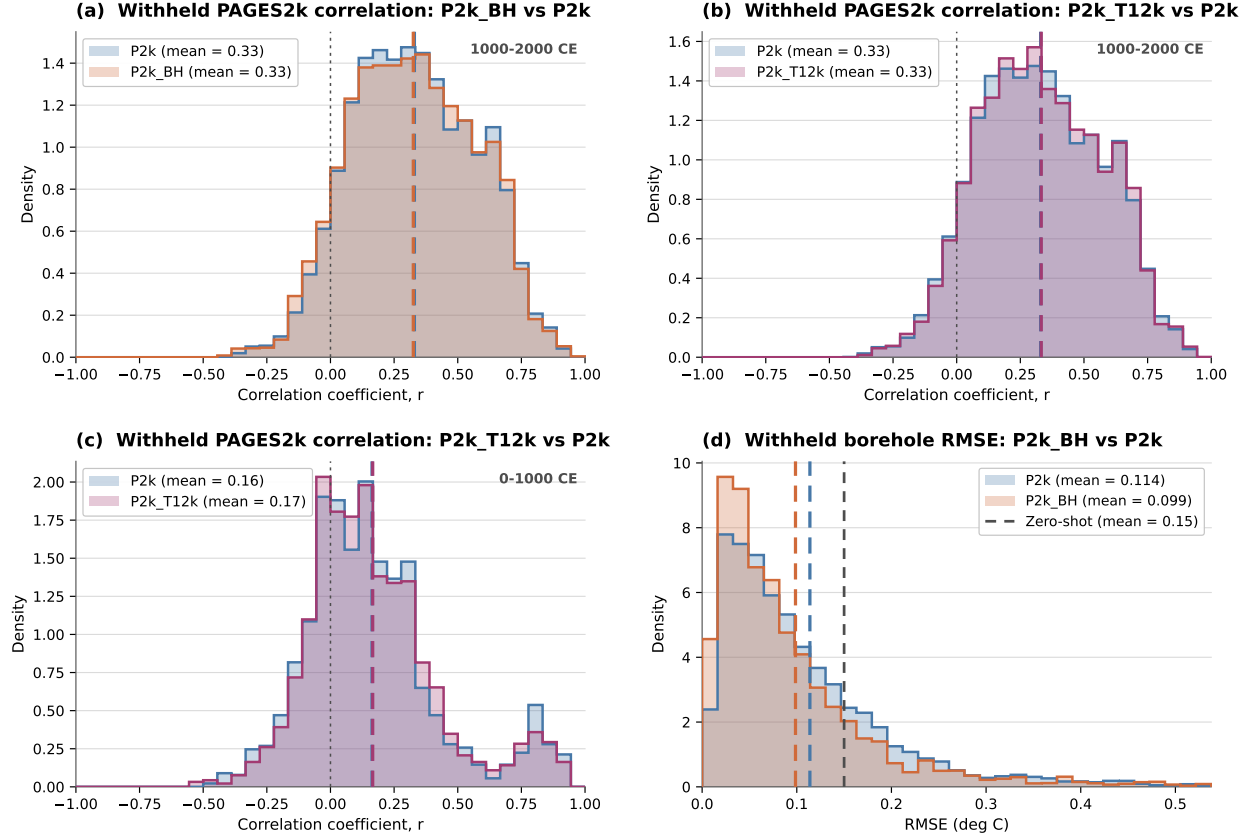


Figure 7: Verification against proxy records withheld from assimilation. Probability-density histograms summarize proxy-level skill across 20 repeated experiments in which 20% of each proxy family was withheld from assimilation. (a) Correlations between observed values of the withheld PAGES2k records and values predicted from the P2k and P2k_BH reconstructions over 1000–2000 CE. (b and c) Corresponding correlations for the P2k and P2k_T12k reconstructions over 1000–2000 CE and 0–1000 CE, respectively. (d) Root-mean-square errors (RMSEs) between withheld borehole temperature profiles and profiles predicted from the P2k and P2k_BH reconstructions; lower RMSE indicates better agreement. The zero-proxy benchmark in (d), labeled “Zero-shot,” is obtained without assimilating any proxy observations. Colored dashed lines indicate the mean of each reconstruction distribution, the gray dashed line in (d) indicates the mean zero-proxy RMSE, and the dotted vertical lines in (a) to (c) mark zero correlation.

Independent verification tests whether the changes produced by the added proxy classes generalize to records excluded from assimilation, rather than reflecting only closer agreement with the records used in the optimization. We randomly withhold 20% of records from each proxy family and repeat the experiments 20 times. Adding boreholes does not degrade correlation skill against withheld PAGES2k records over 1000–2000 CE, and adding Temp12k does not degrade PAGES2k skill over either 1000–2000 CE or 0–1000 CE (Fig. 7). Thus, multiscale assimilation does not sacrifice skill for annually resolved proxies. The strongest improvement occurs in the borehole validation: assimilating PAGES2k already reduces held-out borehole RMSE relative to the zero-proxy case, and adding boreholes further reduces the error.

Finally, the T12k sensitivity experiments test whether the weak-constraint 4D-Var framework can assimilate proxy information across a much broader range of timescales. Temp12k records span decadal to millennial timescales, making them a useful test case for multiscale assimilation beyond the Common Era setting, which here uses proxies at seasonal to annual resolution. In these Temp12k-only reconstructions, the model-error weight parameter J_m controls the balance between the model emulator and the low-frequency proxy constraints (Fig. 8). Because the LIM prior is trained mainly on last-millennium simulations, it does not fully represent Holocene-scale dynamics, particularly the deglaciation during the early Holocene; reducing J_m therefore lowers the effective weight on the model emulator, allowing the optimized trajectory to respond more strongly to the Temp12k observations. As J_m decreases, the reconstructed GMT variability increases, particularly in the early Holocene, while remaining broadly consistent with existing Holocene reconstructions, including the 8.2 ka event and the overall amplitude of Holocene temperature variability. The low- J_m reconstruction

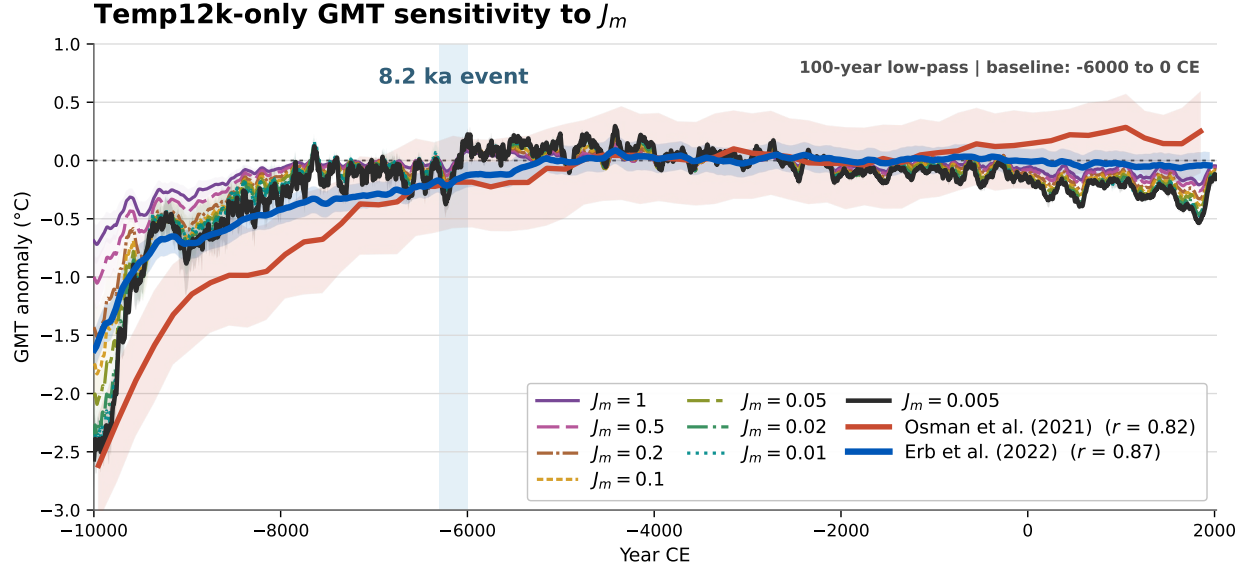


Figure 8: Sensitivity of Temp12k-only global mean temperature reconstructions to the model-error weight J_m . Curves show reconstructed global mean temperature (GMT) anomalies for Temp12k-only experiments with different values of J_m . For each J_m , the plotted curve is the median across the available model-emulator-specific reconstructions, and shading indicates ± 1 standard deviation across model-emulator-specific reconstructions. All series are referenced to the -6000 to 0 CE climatology and smoothed with a 100-year low-pass filter. Temperature reconstructions from [Osman et al. \(2021\)](#) and [Erb et al. \(2022\)](#) are shown for comparison, with shading indicating their reported uncertainty ranges. The 8.2 ka event is highlighted by a light blue band. Correlations are computed separately between the median Temp12k-only reconstruction with $J_m = 0.005$ and the reconstructions of [Osman et al. \(2021\)](#) and [Erb et al. \(2022\)](#) over their respective overlapping intervals.

also shows a mid-Holocene thermal maximum near 6 ka and a late-Holocene cooling trend. This behavior differs from the reconstructions of [Osman et al. \(2021\)](#) and [Erb et al. \(2022\)](#), which do not show the same late-Holocene cooling in the global mean. We therefore interpret the J_m experiments primarily as a sensitivity analysis of the model–data balance, rather than a trustworthy Holocene reconstruction. These results show that LMR4D-Var framework can assimilate proxy records with strongly heterogeneous temporal resolution to yield a 10,000-year-long reconstruction at seasonal resolution. Realistic Holocene-scale applications will however require careful consideration of the model used to perform the optimization.

4 Discussion

The LMR4D-Var framework enables trajectory-level paleoclimate data assimilation across archives that operate on different temporal clocks. This framework combines weak-constraint 4D-Var with a coupled seasonal reduced-order emulator, allowing seasonal, annual, multidecadal, and multicentennial observations to constrain a single evolving climate history. Here, coupling means more than reconstructing several variables simultaneously. Surface air temperature, sea-surface temperature, sea ice, and OHC in the 0–300-m and 300–2000-m layers are represented within one multivariate state and advanced by the same climate-model emulator. Contemporaneous and lagged cross-variable covariances provide the pathway by which observations that directly constrain surface climate can also inform ocean heat storage and sea ice, including regions without assimilated observations. Optimizing the full trajectory propagates these adjustments across variables and through time, while model-error increments allow flexibility in the dynamics that fit the observations. This coupled pathway underlies the recovery of layer-resolved OHC in the pseudo-proxy experiment and the spatially distributed temperature and OHC responses to borehole assimilation.

The OHC reconstruction illustrates that although not directly observed by the proxy network, the coupled state allows surface-temperature-sensitive records to inform slow oceanic variables. The pseudo-proxy experiment supports this pathway under controlled model-world conditions, and the historical trends in both OHC(0–300 m) and OHC(300–2000 m) are consistent with recent ocean-reconstruction and energy-budget estimates, especially [Wu et al. \(2025\)](#). The empirical historical-window trend comparison further indicates that the modern rate of heat accumulation in both

layers lies beyond the reconstructed pre-1870 Common Era distribution. After linear detrending, agreement remains high for OHC(300–2000 m) against Zanna et al. (2019) and Wu et al. (2025), but is lower for OHC(0–300 m) and substantially lower against Gebbie and Huybers (2019). All four experiments track the modern OHC increase, but only those assimilating boreholes achieve high CE against Wu et al. (2025) in the 300–2000-m layer, including after detrending (Fig. 4).

Including terrestrial boreholes also slightly cools LIA global mean temperature but produces a larger coherent OHC response in both layers. This behavior is consistent with the expectation that long-memory archives should be especially informative for slow climate variables. The influence of borehole assimilation is felt globally, and improves validation during the LIA in an independent West Antarctic ice-core borehole record (Orsi et al., 2012). In this sense, LMR4D-Var helps reconcile the cooler low-frequency temperatures often inferred from boreholes with reconstructions based mainly on annually resolved natural proxies (Huang et al., 2000; Council et al., 2007; Christiansen and Ljungqvist, 2017): the archives are treated as complementary constraints on one coupled trajectory rather than as competing estimates of the same instantaneous quantity.

The Temp12k experiments extend the same logic to longer timescales, but they should only be interpreted as a framework-extensibility proof of concept. Temp12k records span decadal to millennial resolutions, so they test whether the framework can assimilate low-frequency temperature information beyond the Common Era. The sensitivity to model error, J_m , shows the importance of allowing for model error in online reconstructions. Lowering the effective weight assigned to the last-millennium-trained emulator allows a larger reconstructed temperature range while retaining consistency with existing Holocene reconstructions.

There are three areas that could further improve upon the results reported here. First, the model emulator is trained on detrended last-millennium simulations in a small state space. This makes coupled seasonal optimization over multicentennial to millennial windows tractable, but comes at the cost of potentially damped multicentennial OHC variability and has no representation of nonlinear ocean adjustment under strong forcing. Moreover, model-error increments cannot restore variability absent from the training simulations. Full Holocene reconstructions will require models that incorporate forcing and a wider range of variability. Second, the observation network primarily constrains surface climate, samples a limited set of archive classes, assumes a diagonal observation-error covariance, and simplifies chronological and temporal-resolution uncertainty. Third, the current state vector excludes precipitation and other hydroclimate variables, limiting the assimilation and reconstruction of Hydro2k and related records. Extending the state vector and differentiable observation operators to precipitation, soil moisture, and other moisture-balance variables would broaden the framework, but would require careful treatment of intermittent and non-Gaussian variability, proxy-system-model uncertainty, and model-dependent hydroclimate covariance.

These caveats notwithstanding, LMR4D-Var provides a flexible platform for reconstructing physically consistent climate trajectories from heterogeneous paleoclimate evidence. The key practical advantage is that any proxy that can be forward-modeled (climate to observation) with an error estimate can be included. This makes the framework well suited for future assimilation of complex proxies whose sensitivities depend on seasonality, archive physics, time averaging, age uncertainty, or the full climate history, including sedimentary proxies, isotope-enabled records, accumulation- or diffusion-affected ice-core signals, and other nonlinear proxy system models (Dolman and Laepple, 2018). The broader contribution is therefore not a single Common Era OHC estimate alone, but a general framework for bringing diverse paleoclimate archives into one reconstruction problem. Beyond reconstruction, this framework may also support long transient simulations. With an appropriate climate model as the dynamical prior, proxy-constrained trajectory optimization could provide better-balanced initial conditions and reduce drift associated with incompletely equilibrated ocean states.

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Author Contributions

Z.M. conceived the study, developed the methodology, performed the analyses, and drafted the manuscript. G.J.H. and E.J.S. supervised the research. G.J.H., E.J.S., and J.E.-G. acquired funding. J.E.-G., E.J.S., and Z.M. contributed to the borehole modeling. Z.M., G.J.H., J.E.-G., E.J.S., and T.G. interpreted the results and revised the manuscript.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Open Research Statement

The plotting and EOF package SACPY (Meng et al., 2023; Meng and Li, 2024) is available at <https://github.com/Zilum/sacpy>. The LMR4D-Var code, experiment configuration files, and processed outputs used to generate the figures in this study are maintained in an active LMR4D-Var development repository. Proxy data products are derived from publicly available source compilations, including PAGES 2k v2.0.0, Temp12k, and published borehole-temperature data sets cited in the main text. The analysis scripts, configuration files, and figure-generation workflow are currently hosted in a private GitHub repository at <https://github.com/Zilum/4DVarLMR>. The repository will be made publicly accessible upon publication and will include documentation of the preprocessing procedures, assimilation settings, and versioned output files needed to reproduce the results reported in this study.

Supporting Information

Additional methods, validation results, 10 supporting figures, and 3 supporting tables are provided in the Supporting Information appended below.

Supporting Information

Supplementary methods, figures, and tables for the main article.

Contents

1. Abbreviations used throughout the manuscript	21
2. Construction of the reduced-order LIM prior	21
3. Weak-constraint 4D-Var in control-variable form	24
4. Borehole forward operator	25
5. Construction of the borehole error covariance	26
6. PAGES2k proxy system models in the reduced LIM space	29
7. Temp12k as a direct low-frequency temperature target	29
8. Trend significance test	29
9. Additional validation and reconstruction diagnostics	30

List of Tables

Table S1. Common abbreviations used in the manuscript	21
Table S2. Model simulations used to train the LIM priors	22
Table S3. Reconstruction experiments analyzed in the main text	22

List of Figures

Figure S1. Pseudo-proxy comparison with the target CESM-LME simulation	23
Figure S2. Example construction of borehole observation error	28
Figure S3. Spatial OHC correlations with Wu et al. (2025)	31
Figure S4. Comparison of 1880–2000 CE OHC trends with Wu et al. (2025)	32
Figure S5. Global-mean OHC(300–2000 m) comparisons across experiments	33
Figure S6. Layer-resolved OHC comparison with an external reconstruction	34
Figure S7. Layer-resolved OHC changes in the model simulations	35
Figure S8. OHC trends across the MWP-to-LIA transition	36
Figure S9. Pairwise reconstruction comparisons, part A	37
Figure S10. Pairwise reconstruction comparisons, part B	38

1. Abbreviations used throughout the manuscript

Table S1 summarizes the abbreviations used repeatedly in the main text and supporting information. After first definition in the main text, these abbreviations are used consistently throughout the manuscript.

Table S1: Common abbreviations used in the manuscript.

Abbreviation	Meaning
LMR4D-Var	Last Millennium Reanalysis 4D-Var Framework
LIM	Linear Inverse Model
PSM	Proxy system model
CE	Coefficient of Efficiency
GMT	Global mean temperature
OHC	Ocean heat content
OHC0–300	Ocean heat content integrated over 0–300 m
OHC300–2000	Ocean heat content integrated over 300–2000 m
OHC0–2000	Ocean heat content integrated over 0–2000 m
TAS	Near-surface air temperature
TOS	Sea-surface temperature
SIT	Northern Hemisphere sea-ice thickness
SIC	Northern Hemisphere sea-ice concentration
RWP	Roman Warm Period
MWP	Medieval Warm Period
LIA	Little Ice Age

2. Construction of the reduced-order LIM prior

The reduced-order dynamical prior is represented by a discrete Linear Inverse Model (LIM),

$$\mathbf{x}_{k+1} = \mathbf{G} \mathbf{x}_k + \mathbf{n}_k, \quad (\text{S1})$$

where $\mathbf{x}_k$ is the reduced climate state at seasonal analysis step k , $\mathbf{G}$ is the one-step propagator, and $\mathbf{n}_k$ is an additive model-error increment. Following standard LIM estimation, $\mathbf{G}$ is obtained from the lag-0 and lag- τ covariance matrices of the training trajectory (Penland and Magorian, 1993b):

$$\mathbf{G} = \mathbf{C}(\tau)\mathbf{C}(0)^{-1}. \quad (\text{S2})$$

Assuming a stationary training process and model-error increments that are uncorrelated with the preceding state, the corresponding discrete model-error covariance is

$$\mathbf{Q} = \mathbf{C}(0) - \mathbf{G}\mathbf{C}(0)\mathbf{G}^T. \quad (\text{S3})$$

Thus $\mathbf{n}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q})$. The increments are assumed to be temporally uncorrelated between seasonal steps, while $\mathbf{Q}$ retains the full covariance among reduced-state components within each step. Equations (S1)–(S3) fully specify the discrete LIM prior used by LMR4D-Var; the construction of its reduced state and training data is described next.

The LIM is trained in a reduced EOF space built from coupled climate-model integrations. In the present study the state vector contains principal components for six variable groups: 2-m air temperature (TAS), sea-surface temperature (TOS), ocean heat content in the upper 300 m (OHC0–300), ocean heat content integrated over 300–2000 m (OHC300–2000), Northern Hemisphere sea-ice thickness (SIT), and Northern Hemisphere sea-ice concentration (SIC). We use the first 15 EOF modes for each variable except for OHC0–300 and OHC300–2000, for which we use the first 30 modes to capture the more complex spatial structure of upper-ocean heat content and low-frequency variability in the ocean interior. The total dimension of the reduced state is therefore 120. The reduced state is written schematically as

$$\mathbf{x} = [\mathbf{PC}_{\text{TAS}}^T, \mathbf{PC}_{\text{TOS}}^T, \mathbf{PC}_{\text{OHC0-300}}^T, \mathbf{PC}_{\text{OHC300-2000}}^T, \mathbf{PC}_{\text{SIT}}^T, \mathbf{PC}_{\text{SIC}}^T]^T. \quad (\text{S4})$$

Upper-2000-m ocean heat content is then diagnosed as

$$\text{OHC}_{0-2000} = \text{OHC}_{0-300} + \text{OHC}_{300-2000}. \quad (\text{S5})$$

Table S2: Model simulations used to train the LIM priors. Each model family is used to estimate lagged covariance statistics and train a separate reduced-order prior.

Model source	Model name	Experiment class	Simulation period	Role in LMR4D-Var
CESM LME	CESM Last Millennium Ensemble	CESM millennium ensemble	850–1850 CE	Multimember coupled prior
PMIP3	MPI-ESM-P	Last millennium simulation	850–1850 CE	Single-model LIM prior
PMIP4	MPI-ESM1.2-LR	Last millennium simulation	850–1850 CE	Single-model LIM prior
PMIP4	MRI-ESM2-0	Last millennium simulation	0–1850 CE	Single-model LIM prior
PMIP3	CCSM4	Last millennium simulation	850–1850 CE	Single-model LIM prior

Table S3: Summary of the five reconstruction experiments analyzed in the main text. The experiment design isolates the effect of adding low-frequency and long-memory proxy information and includes a Temp12k-only experiment for Holocene-scale sensitivity tests.

Experiment	PAGES2k	Temp12k	Boreholes	Purpose
P2k	Yes	No	No	Baseline Common Era reconstruction
P2k_BH	Yes	No	Yes	Isolate borehole long-memory constraints
P2k_T12k	Yes	Yes	No	Isolate Temp12k low-frequency constraints
P2k_BH_T12k	Yes	Yes	Yes	Combined multiscale reconstruction
T12k	No	Yes	No	Temp12k-only Holocene and J_m sensitivity tests

Here, OHC_{0-300} and $\text{OHC}_{300-2000}$ are diagnosed from the respective EOF modes. This partition is useful because it allows the LIM to represent the fast upper-ocean component and the slower subsurface reservoir separately while still recovering the full upper-2000-m quantity analyzed in the main text.

Each variable is area weighted before EOF analysis. EOF truncation is applied separately to each variable class to limit the dimension of the coupled state while retaining the leading variance and covariance structures needed for prediction. In practice this truncation is a compromise: too few modes fail to capture important coupled variability, whereas too many modes can degrade skill and overfit noise. The resulting EOF-based state vectors are assembled from seasonal-mean fields and used to estimate lagged covariance statistics for each training ensemble.

Before the LIM is estimated, each parent simulation is linearly detrended over its training interval after remapping and seasonal averaging. This step removes slow unrealistically persistent adjustment associated with climate-model drift, or long equilibration of the ocean background state. If such behavior were left in the training data, the fitted LIM operator would absorb that nonstationary drift into its linear dynamics, which would in turn bias the propagator toward unstable or physically misleading low-frequency tendencies. The detrending step is therefore intended to isolate internally covarying seasonal-to-multidecadal variability rather than the model’s particular centennial-scale drift history.

The LIMs are then trained from multiple coupled climate-model priors, including the CESM Last Millennium Ensemble and PMIP-class millennium simulations from MPI-ESM-P, MPI-ESM1.2-LR, MRI-ESM2-0, and CCSM4, after remapping to a common grid, forming seasonal means, and detrending the training trajectories. Training separate LIM priors from multiple parent simulations allows the reconstruction system to sample a broader range of coupled covariance structures and reduces dependence on any one model’s drift history or initial condition. Table S2 summarizes the training simulations, and Table S3 summarizes the proxy composition and purpose of the reconstruction experiments.

As a controlled evaluation of the coupled reconstruction, we conduct a pseudo-proxy experiment using CESM-LME member 001 as the target trajectory. Proxy values are generated from the target simulation using the same proxy locations and observation operators used in the Common Era reconstruction. We adopt a leave-one-model-family-out design in which the entire CESM-LME ensemble, rather than only the target member, is excluded from construction of the EOF basis and LIM priors. The target trajectory is reconstructed using the four model-specific LIM priors trained from the remaining climate-model ensembles. This design tests whether the weak-constraint framework can recover known variations in GMT, OHC_{0-300} , and $\text{OHC}_{300-2000}$ from sparse surface-temperature-sensitive proxy information without using dynamical information from the target model family. The resulting comparison is shown in Fig. S1. OHC series in this diagnostic are linearly detrended before anomaly calculation so that the comparison emphasizes reconstructed variability and covariance structure rather than differences in background model drift.

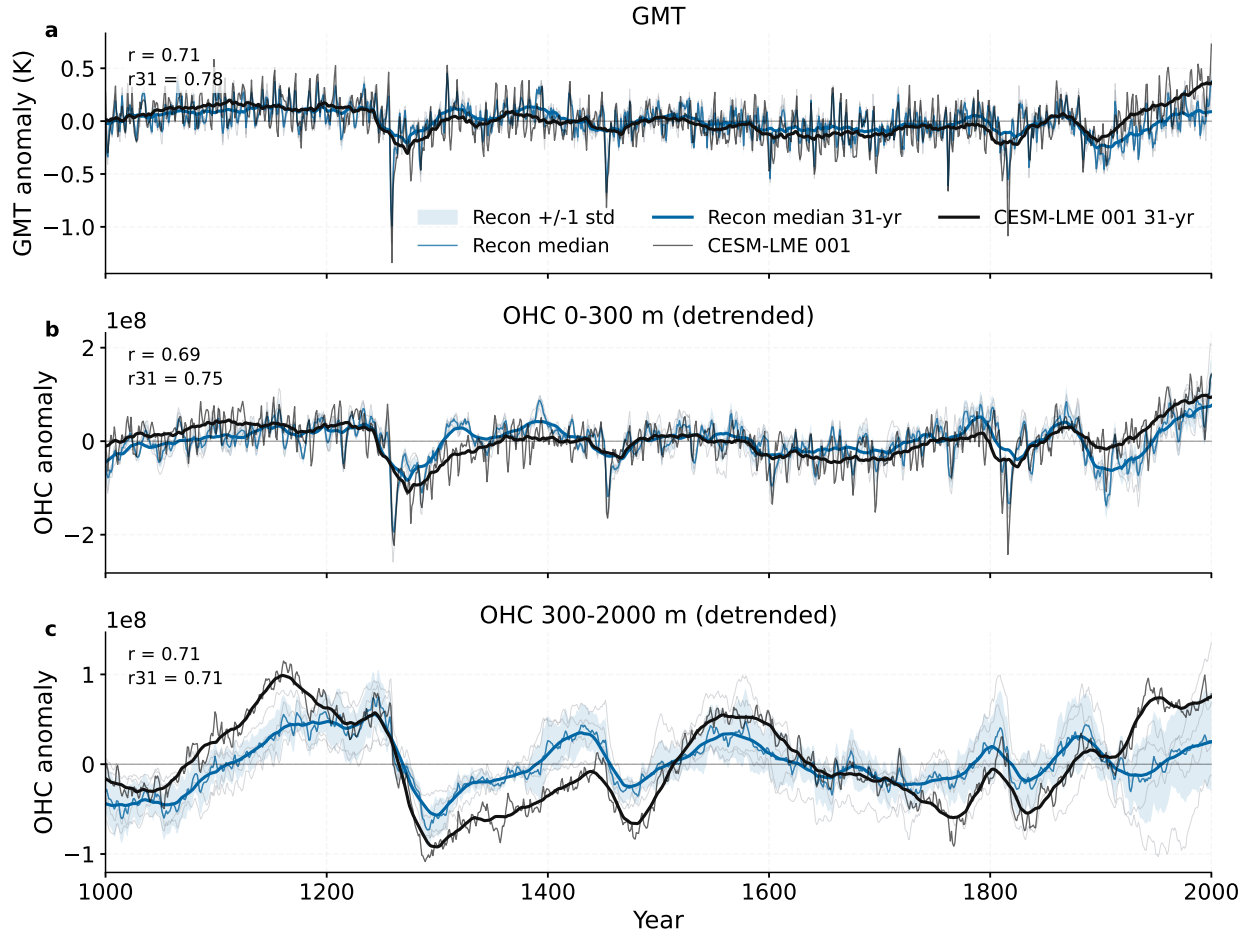


Figure S1: Pseudo-proxy experiment comparison between four prior-specific reconstructions and the target CESM-LME member 001. Panels show annual anomalies over 1000–2000 CE for (a) global mean temperature (GMT), (b) global ocean heat content from 0–300 m, and (c) global ocean heat content from 300–2000 m. All CESM-LME simulations are excluded from the EOF basis and LIM-prior construction, and the target is reconstructed using the four remaining model-specific priors. Blue lines show the median across the four prior-specific reconstructions, with light blue shading indicating ± 1 inter-prior standard deviation. Thin gray lines show the individual model-emulator-specific reconstructions, and black lines show CESM-LME member 001. Thick blue and black lines indicate 31-year running means. OHC series are linearly detrended before anomaly calculation. Correlations between the annual and 31-year-smoothed reconstruction median and CESM-LME member 001 are shown in each panel.

3. Weak-constraint 4D-Var in control-variable form

Four-dimensional variational data assimilation estimates a model trajectory by minimizing its mismatch with observations distributed across an assimilation window (Le Dimet and Talagrand, 1986; Talagrand and Courtier, 1987; Courtier et al., 1994; Rabier et al., 2000). Weak-constraint 4D-Var extends this formulation by allowing departures from the model dynamics through explicit model-error increments, rather than assuming a perfect model over the full window (Sasaki, 1970; Fisher et al., 2005; Trémolet, 2006; Fisher et al., 2011; Evensen et al., 2022). Following the forcing or control-variable formulation of weak-constraint 4D-Var (Trémolet, 2006; Fisher et al., 2011), the unknown initial state and the sequence of seasonal model-error increments for a window containing states $\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_N$ are collected into the control vector

$$\mathbf{v} = \begin{pmatrix} \mathbf{x}_0 \\ \mathbf{n}_0 \\ \mathbf{n}_1 \\ \vdots \\ \mathbf{n}_{N-1} \end{pmatrix}. \quad (\text{S6})$$

Applying Equation (S1) recursively gives the state at seasonal step k as

$$\mathbf{x}_k = \mathbf{G}^k \mathbf{x}_0 + \sum_{j=0}^{k-1} \mathbf{G}^{k-1-j} \mathbf{n}_j, \quad k = 1, \dots, N. \quad (\text{S7})$$

Equation (S7) shows that each state depends on the initial condition and on every model-error increment from the preceding seasonal steps. An increment $\mathbf{n}_j$ therefore modifies the state at step $j + 1$ and propagates through $\mathbf{G}$ to all subsequent states. Collecting these relations for the full assimilation window gives a linear transformation from control space to trajectory space (Fisher et al., 2005, 2011; Evensen et al., 2022):

$$\mathbf{x} = \mathbf{U}\mathbf{v}, \quad (\text{S8})$$

where $\mathbf{x} = (\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_N)^T$ and $\mathbf{U}$ is the lower block-triangular matrix

$$\mathbf{U} = \begin{pmatrix} \mathbf{I} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{G} & \mathbf{I} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{G}^2 & \mathbf{G} & \mathbf{I} & \cdots & \mathbf{0} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mathbf{G}^N & \mathbf{G}^{N-1} & \mathbf{G}^{N-2} & \cdots & \mathbf{I} \end{pmatrix}. \quad (\text{S9})$$

The initial state is assumed to be uncorrelated with future model-error increments, and model-error increments are assumed to be uncorrelated between seasonal steps. Their covariance within each step is nevertheless fully represented by $\mathbf{Q}$, allowing spatial and cross-variable covariance among the reduced-state components. Under these commonly used weak-constraint assumptions (Trémolet, 2006; Fisher et al., 2011; Evensen et al., 2022), the control-space prior covariance is block diagonal:

$$\mathbf{B}_v = \begin{pmatrix} \mathbf{C}(0) & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{Q} & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{Q} \end{pmatrix}, \quad (\text{S10})$$

Here $\mathbf{C}(0)$ is the stationary covariance of the initial reduced state and $\mathbf{Q}$ is defined by Equation (S3). Assuming Gaussian prior, model, and observation errors, the weak-constraint cost function follows the standard quadratic variational form (Le Dimet and Talagrand, 1986; Fisher et al., 2005; Trémolet, 2006; Evensen et al., 2022):

$$J(\mathbf{v}) = \frac{1}{2} \mathbf{v}^T \mathbf{B}_v^{-1} \mathbf{v} + \frac{1}{2} [\mathbf{y} - \mathcal{H}(\mathbf{U}\mathbf{v})]^T \mathbf{R}^{-1} [\mathbf{y} - \mathcal{H}(\mathbf{U}\mathbf{v})].$$

Expanding the control-space prior term separates the initial-state and model-error contributions:

$$J(\mathbf{v}) = \underbrace{\frac{1}{2} \mathbf{x}_0^T \mathbf{C}(0)^{-1} \mathbf{x}_0}_{\text{initial-state constraint}} + \underbrace{\frac{1}{2} \sum_{k=0}^{N-1} \mathbf{n}_k^T \mathbf{Q}^{-1} \mathbf{n}_k}_{\text{model-error penalty}} + \underbrace{\frac{1}{2} [\mathbf{y} - \mathcal{H}(\mathbf{U}\mathbf{v})]^T \mathbf{R}^{-1} [\mathbf{y} - \mathcal{H}(\mathbf{U}\mathbf{v})]}_{\text{proxy observation misfit}}. \quad (\text{S11})$$

The three terms constrain the initial state, departures from the LIM trajectory, and mismatch to the proxy observations, respectively. Here $\mathcal{H}$ contains all proxy-specific and history-dependent observation operators, and $\mathbf{R}$ is the observation-error covariance. The control-space formulation avoids constructing a dense covariance for the full trajectory while allowing observations with long temporal footprints, such as boreholes, to constrain multiple analysis times (Fisher et al., 2005, 2011). After optimization to get $\mathbf{v}^*$, the analyzed trajectory is $\mathbf{x}_a = \mathbf{U}\mathbf{v}^*$.

4. Borehole forward operator: from surface temperature history to subsurface profile

The borehole observation operator links a reconstructed ground-surface temperature history to the present-day temperature anomaly profile measured in the subsurface. Following standard borehole climatology theory (Huang et al., 2000; Cuesta-Valero et al., 2021), the continental subsurface is treated as a one-dimensional semi-infinite conductive medium with depth coordinate z positive downward. The observed temperature profile is written as the sum of a long-term quasi-equilibrium profile and a transient perturbation,

$$T(z) = T_0 + \Gamma_0 z + T_t(z), \quad (\text{S12})$$

where T_0 is the long-term surface temperature, Γ_0 is the background geothermal gradient, and $T_t(z)$ is the transient anomaly induced by past changes in surface temperature.

For a homogeneous medium without internal heat sources, the transient temperature evolves according to the one-dimensional heat-diffusion equation

$$\frac{\partial T}{\partial t} = \kappa \frac{\partial^2 T}{\partial z^2}, \quad (\text{S13})$$

where κ is thermal diffusivity. An instantaneous surface step change propagates downward as a complementary-error-function profile, so a discretized surface-temperature history can be represented as a superposition of step responses. This solution also implies a characteristic diffusion timescale

$$t \sim \frac{Z^2}{4\kappa},$$

obtained by setting the similarity variable $\eta = z/(2\sqrt{\kappa t})$ to order one. The relationship shows that the response time grows quadratically with depth: for $\kappa \sim 10^{-6} \text{ m}^2 \text{ s}^{-1}$, temperatures at 100 m depth primarily reflect variability on the order of many decades to about a century, with deeper levels integrating still longer timescales. If the surface-temperature history is sampled at discrete times and written as $\mathbf{T}_s = (T_s(t_1), T_s(t_2), \dots, T_s(t_{N_t}))^T$, then the transient anomaly at depth z_i can be written as

$$T_t(z_i) = \sum_{j=1}^{N_t} T_s(t_j) \left[\text{erfc}\left(\frac{z_i}{2\sqrt{\kappa t_j}}\right) - \text{erfc}\left(\frac{z_i}{2\sqrt{\kappa t_{j-1}}}\right) \right]. \quad (\text{S14})$$

This is the forward model that maps a surface-temperature history to a borehole temperature anomaly profile.

In matrix form, the discrete borehole operator can be written as

$$\mathbf{y}_{\text{bh}} = \mathbf{M}\mathbf{T}_s, \quad (\text{S15})$$

where $\mathbf{T}_s$ is the discretized surface-temperature history and $\mathbf{y}_{\text{bh}}$ is the predicted borehole temperature anomaly profile at the observed depths. In other words, the borehole forward model in LMR4D-Var maps surface temperature through time directly to subsurface temperature through depth, $\mathbf{T}_s \mapsto \mathbf{y}_{\text{bh}}$. The elements of $\mathbf{M}$ are

$$M_{ij} = \text{erfc}\left(\frac{z_i}{2\sqrt{\kappa t_j}}\right) - \text{erfc}\left(\frac{z_i}{2\sqrt{\kappa t_{j-1}}}\right). \quad (\text{S16})$$

Within LMR4D-Var, $\mathbf{T}_s$ is supplied by the reconstructed near-surface temperature history at each borehole location, and the resulting profile is compared with the observed borehole anomaly profile in the observation-misfit term of the weak-constraint 4D-Var objective. In this way the borehole proxy constrains the low-frequency integrated thermal history of the land surface rather than the instantaneous climate state at a single analysis time. Because the purpose of the borehole constraint here is to isolate long-timescale information, only the part of each borehole profile below 100 m is used in the forward comparison and subsequent error construction.

5. Construction of the borehole error covariance

The borehole observation error used in the present study is intended as an effective error model rather than as a pure estimate of measurement noise alone. In practice, borehole profiles can be affected by site-specific environmental disturbances and observational complications, including groundwater flow, snow and surface-condition effects, and other departures from ideal one-dimensional conductive behavior. A second source of uncertainty arises from the forward model itself: LMR4D-Var uses a simplified conductive diffusion model to simulate borehole temperatures, and any variance not captured by that model must also be absorbed into the observation-error description. Rather than assigning a depth-independent error or a single site-wide scalar uncertainty, we therefore estimate a depth-dependent residual variance for each borehole profile and use that quantity to define the diagonal borehole error covariance. Fig. S2 summarizes this procedure panel by panel.

The construction begins from the raw borehole temperature profile $T_{\text{raw}}(z)$ shown in panel A. Following the geothermal treatment used in borehole climatology studies (Cuesta-Valero et al., 2021), we estimate the quasi-equilibrium geothermal profile by fitting a straight line over the 200–300 m interval,

$$T_{\text{geo}}(z) = a + bz,$$

where the fitted slope b defines the local geothermal gradient. Uncertainty in the fitted slope is propagated to construct a mean geothermal fit together with upper and lower $\pm 2\sigma$ fits, also shown in panel A. Subtracting these fits from the raw profile yields the perturbation profiles shown in panel B,

$$T_{\text{pert}}(z) = T_{\text{raw}}(z) - T_{\text{geo}}(z),$$

including the mean perturbation profile and the perturbation envelope associated with uncertainty in the background geothermal trend.

Each perturbation profile is then inverted to obtain a candidate ground-surface temperature history, as illustrated in panel C, using the same inverse geothermal reconstruction framework as in Cuesta-Valero et al. (2021). In that framework the perturbation profile is written in matrix form as

$$\mathbf{y}_{\text{bh}} = \mathbf{M}\mathbf{T}_s,$$

and the inverse problem is solved with singular value decomposition (SVD). If

$$\mathbf{M} = \mathbf{U}\mathbf{S}\mathbf{V}^T,$$

then the candidate surface-temperature history is obtained from the truncated pseudo-inverse,

$$\hat{\mathbf{T}}_s^{(k)} = \mathbf{V}\mathbf{S}_k^{-1}\mathbf{U}^T\mathbf{y}_{\text{bh}},$$

where $\mathbf{S}_k^{-1}$ retains only the leading k singular values and sets the smaller ones to zero. Writing the inverse explicitly in this way is useful because the truncation level is one of the ensemble dimensions used to characterize borehole uncertainty.

Following Cuesta-Valero et al. (2021), we consider step-change models with 25-, 40-, and 50-year temporal resolution, corresponding to 16, 10, and 8 time steps over the reconstructed interval. We also allow the thermal properties entering the conductive kernel to vary across a broad but physically plausible range. Specifically, the inversion ensemble spans thermal conductivities of 2.5, 3.0, and 3.5 W m⁻¹ K⁻¹ and volumetric heat capacities of 2.5, 3.0, and 3.5 × 10⁶ J m⁻³ K⁻¹, which together yield nine thermal diffusivities ranging from approximately 0.7 × 10⁻⁶ to 1.4 × 10⁻⁶ m² s⁻¹. We use the same SVD truncation strategy: the 25-year model retains the leading 3, 4, or 5 singular values; the 40-year model retains the leading 2, 3, or 4 singular values; and the 50-year model also retains the leading 2, 3, or 4 singular values. These alternative truncations suppress unrealistically amplified small-eigenvalue solutions while still allowing a broad family of admissible inverse histories.

The total number of candidate histories generated from a single original borehole profile is therefore

$$N_{\text{cand}} = 3 \times 9 \times 3 \times 3 = 243,$$

where the four factors correspond to the three perturbation profiles from panel B, the nine thermal-property combinations, the three step-change models, and the three SVD truncation choices available within each step-change model. The borehole inversion in panel C should therefore be interpreted not as a single reconstruction but as a 243-member inverse ensemble for each site.

In the present implementation, the candidate histories obtained from this large inversion ensemble are then filtered, as shown in panel C, to remove solutions with implausibly large temporal gradients exceeding 4 K per decade. This step is

intended to discard unstable inverse solutions that fit the perturbation profile only by introducing unrealistically abrupt or high-magnitude temperature changes. The retained ensemble therefore represents the subset of inverse solutions that remain physically plausible while still reproducing the observed perturbation structure.

The retained temperature histories are then projected forward again through the borehole operator to reconstruct perturbation profiles in depth space, producing the curves shown in panel D:

$$\hat{\mathbf{y}}_{\text{bh}} = \mathbf{M}\hat{\mathbf{T}}_s,$$

where $\hat{\mathbf{T}}_s$ denotes a retained inverse temperature history. These forward-reconstructed perturbation profiles are compared with the original perturbation profile from panel B to form a depth-dependent residual ensemble. Let $r_i^{(m)}$ denote the residual at depth z_i for retained solution m ,

$$r_i^{(m)} = \hat{y}_{\text{bh},i}^{(m)} - y_{\text{bh},i}^{\text{orig}}.$$

Panel E summarizes the resulting depth-dependent residual variance, approximated by the mean squared residual across the retained ensemble,

$$\sigma_{\text{bh}}^2(z_i) = \frac{1}{N_m} \sum_{m=1}^{N_m} \left(r_i^{(m)} \right)^2, \quad (\text{S17})$$

which is then used to define the site-level diagonal borehole observation-error covariance,

$$\mathbf{R}_{\text{bh}} = \text{diag} \left(\sigma_{\text{bh}}^2(z_1), \sigma_{\text{bh}}^2(z_2), \dots, \sigma_{\text{bh}}^2(z_{N_z}) \right). \quad (\text{S18})$$

This construction allows the borehole uncertainty to vary with depth, so that levels where the inverse-plus-forward reconstruction is less stable carry larger observational uncertainty than levels where the reconstruction is more tightly constrained.

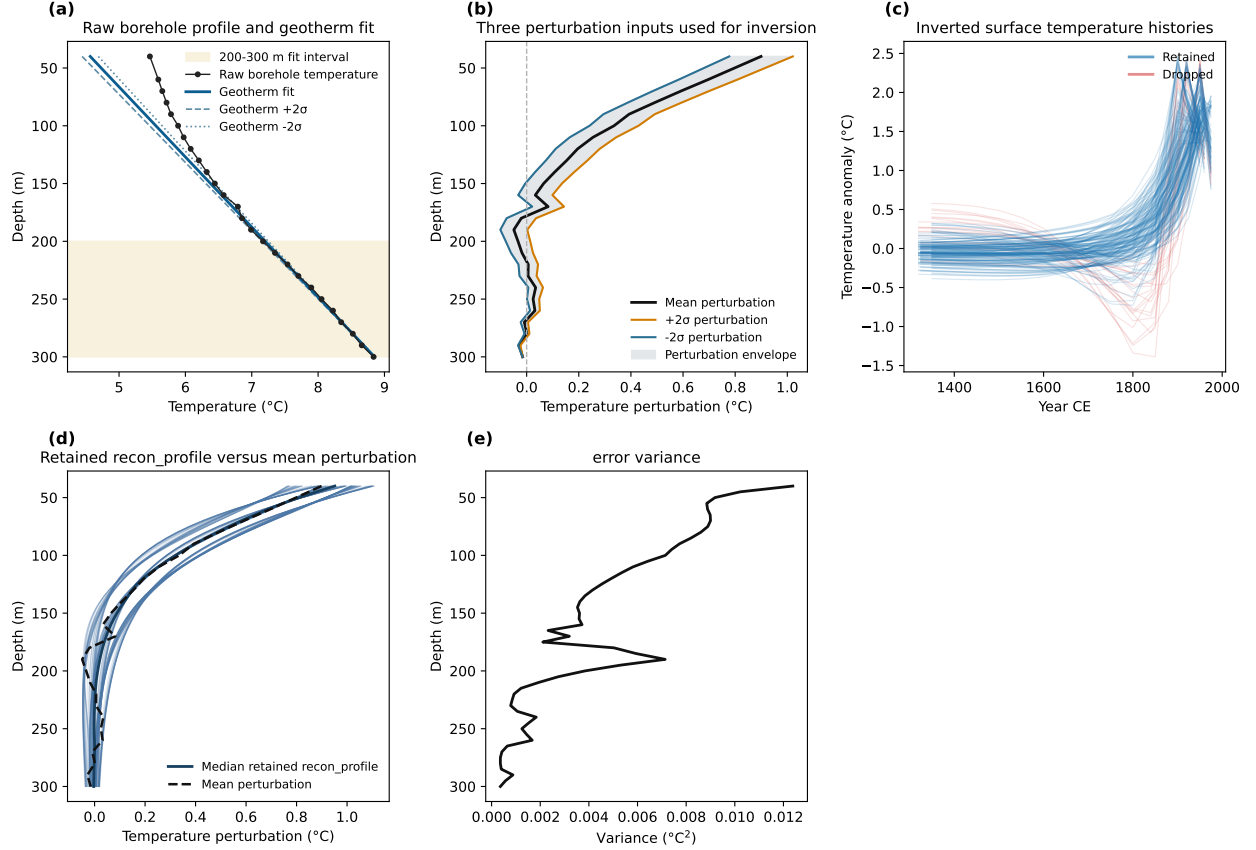


Figure S2: Example construction of borehole error for station CA_0011. (a) Raw borehole temperature profile (black) as a function of depth, together with the mean geothermal fit estimated from the 200–300 m interval (blue) and the corresponding $\pm 2\sigma$ geothermal fits (dashed and dotted lines) derived from uncertainty in the fitted gradient. (b) Temperature perturbation profiles obtained by subtracting the mean and $\pm 2\sigma$ geothermal fits from the raw borehole temperature profile. These three perturbation estimates define the perturbation envelope associated with uncertainty in the background geothermal trend. (c) Surface-temperature histories inferred from the perturbation profiles using the same inverse geothermal reconstruction method as in Section 5. (d) Perturbation profiles reconstructed by forward-projecting the retained temperature histories back into depth space. (e) Depth-dependent variance of the residual between the forward-reconstructed perturbation profiles and the original perturbation profile. This residual variance is used to approximate the borehole observation-error covariance, with the diagonal borehole error matrix constructed from the depthwise mean squared residuals.

6. PAGES2k proxy system models in the reduced LIM space

The annually resolved baseline network uses the temperature-sensitive PAGES2k phase 2 proxy compilation (PAGES2k Consortium and others, 2017). These records include tree rings, corals, ice cores, and other annually to seasonally resolved archives. In LMR4D-Var they are assimilated through simple temperature-sensitive proxy system models (PSMs) calibrated against instrumental target fields rather than against the full unreduced climate state.

Calibration is performed separately for terrestrial and marine records. Terrestrial proxies are calibrated against instrumental near-surface air temperature from GISTEMP v4 (Lenssen et al., 2019), whereas marine proxies are calibrated against instrumental sea-surface temperature from ERSST v5 (Huang et al., 2017). Moisture variables are not included in the PSM calibration. This choice follows the temperature-centered design of the present reconstruction and avoids introducing additional predictor fields that are not represented uniformly across the reduced LIM state.

Because the LIM evolves a truncated EOF representation of the coupled climate state rather than the full local gridpoint climate field, the PSMs are calibrated in the same reduced space used by the assimilation. Let $\mathbf{x}$ denote the reduced LIM state and let $\hat{\mathbf{x}}$ denote its projection back onto the truncated climate subspace. Then

$$\hat{\mathbf{x}} = \mathbf{E}\mathbf{E}^T \mathbf{x}, \quad (\text{S19})$$

where $\mathbf{E}$ denotes the EOF basis matrix for the relevant climate field. The linear PSM for an individual record is written as

$$\mathbf{y} = \mathbf{H}\hat{\mathbf{x}} + \boldsymbol{\epsilon}, \quad (\text{S20})$$

where $\mathbf{H}$ maps the truncated climate field to the proxy estimate and $\boldsymbol{\epsilon}$ is the residual error term. Calibrating in the EOF-truncated space partially absorbs representativeness error associated with unresolved local climate structure.

Observation-error statistics are represented with a diagonal covariance matrix, so that

$$\mathbf{R} = \langle \boldsymbol{\epsilon}\boldsymbol{\epsilon}^T \rangle \quad (\text{S21})$$

is assumed to have negligible off-diagonal covariance among distinct PAGES2k records. This approximation is adopted because the calibration interval is too short and unevenly overlapping to estimate a stable full proxy-error covariance matrix, and exploratory tests indicate that the diagonal error variances dominate the off-diagonal terms.

Seasonality is determined objectively during calibration by selecting the seasonal target temperature that gives the best instrumental fit, following the logic used in previous LMR implementations. Sub-seasonal coral records are first aggregated to seasonal resolution before calibration and assimilation. Records are screened before use in the assimilation: proxies with very weak local temperature calibration or strongly autocorrelated residuals are excluded so that the assimilated network remains consistent with the assumption of temporally uncorrelated observation errors at the annual analysis scale.

7. Temp12k as a direct low-frequency temperature target

The Temp12k compilation is not treated through an additional calibrated proxy system model. Instead, the compilation is assimilated directly because it already provides reconstructed temperature anomalies together with uncertainty estimates in temperature units (Kaufman et al., 2020). In that sense, Temp12k enters the variational objective as a low-frequency temperature target whose reported anomaly and uncertainty replace the role that a separate PSM calibration would otherwise play. To maintain consistency with the Common Era reconstruction shown in the main text, Temp12k anomalies are referenced to the 0–2000 CE climatology before assimilation.

8. Trend significance test

This section describes the significance test used for the spatial trend maps in Figs. S4 and S8. At each grid point, we estimate the linear trend in annual OHC anomalies for each depth layer with an ordinary least squares regression, following standard practice for climate-trend estimation (Santer et al., 2000; Wilks, 2011),

$$y(t) = a + bt + \varepsilon(t),$$

where $y(t)$ is the annual OHC anomaly, a is the intercept, b is the linear trend, and $\varepsilon(t)$ is the residual. The plotted trend is the slope estimate $\hat{b}$, reported in $\text{J m}^{-2} \text{ century}^{-1}$.

We test the null hypothesis $H_0 : b = 0$ against the two-sided alternative $H_1 : b \neq 0$. Because annual OHC anomalies are serially correlated, the nominal sample size overstates the effective degrees of freedom. We therefore estimate the lag-1 autocorrelation r_1 of the regression residuals and use it to define the effective sample size

$$n_{\text{eff}} = n \frac{1 - r_1}{1 + r_1},$$

where n is the number of annual observations. This lag-1 adjustment is a standard approximation for climate time series with red-noise persistence (Santer et al., 2000; Wilks, 2011). The effective degrees of freedom are then taken as $\text{df} = n_{\text{eff}} - 2$.

Using this adjusted sample size, we estimate the residual variance and the standard error of the slope in the usual way,

$$\text{MSE} = \frac{\sum_{i=1}^n \varepsilon_i^2}{\text{df}}, \quad \text{SE}(\hat{b}) = \sqrt{\frac{\text{MSE}}{\sum_{i=1}^n (t_i - \bar{t})^2}},$$

and compute the corresponding t -statistic,

$$t = \frac{\hat{b}}{\text{SE}(\hat{b})}.$$

The two-sided p -value is obtained from the Student's t distribution with df degrees of freedom. We account for simultaneous testing across each spatial field using the Benjamini-Hochberg procedure, with the false discovery rate (FDR) controlled at $q = 0.05$ (Benjamini and Hochberg, 1995; Wilks, 2016). For each panel, the m valid ocean-grid-cell p -values are ordered as

$$p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(m)}.$$

We identify the largest rank k satisfying

$$p_{(k)} \leq \frac{k}{m} q$$

and define the panel-specific threshold as $p_{\text{FDR}}^* = p_{(k)}$. Grid cells with $p_i \leq p_{\text{FDR}}^*$ are classified as FDR significant. Thus, $q = 0.05$ controls the expected proportion of false discoveries among the grid cells classified as significant, rather than the probability that any individual grid cell is a false positive.

The correction is applied separately to all valid ocean grid cells in each of the six panels in Figs. S4 and S8. Land and invalid grid cells are excluded and remain missing in the trend, raw- p , FDR-adjusted q , and significance-mask fields. The effective-sample-size adjustment addresses temporal autocorrelation, whereas the panel-wise BH-FDR procedure addresses spatial multiple testing. In the figures, unmasked colored regions pass the FDR-controlled test, semitransparent gray overlays identify regions that do not pass, and land or invalid grid cells remain blank.

9. Additional validation and reconstruction diagnostics

The following figures provide spatial and layer-resolved comparisons of reconstructed OHC with external estimates, document the OHC behavior of the model simulations used to train the emulators, compare historical OHC changes across selected intervals, and show pairwise comparisons among the proxy-assimilation experiments.

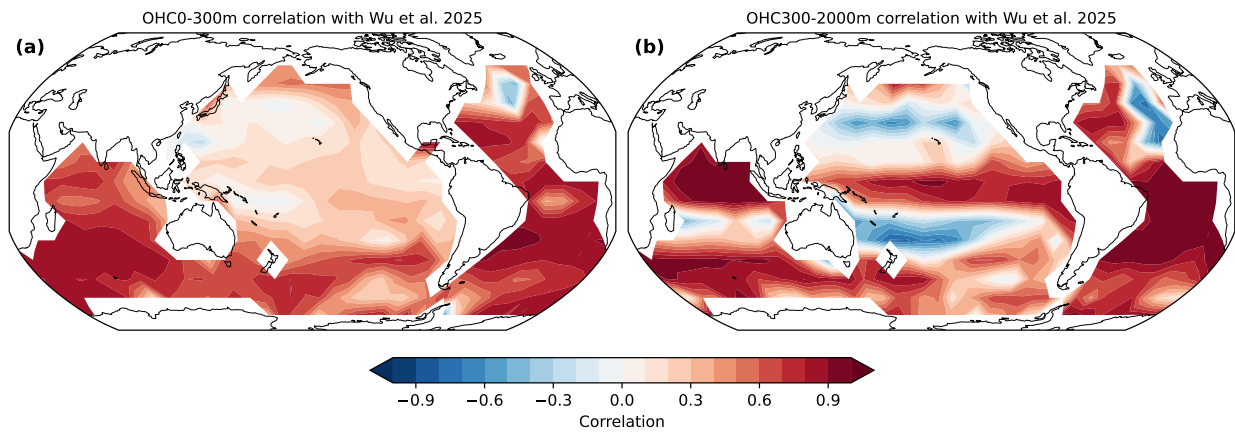


Figure S3: Spatial correlation between reconstructed ocean heat content and Wu et al. (2025). Maps show gridpoint correlations between the ocean heat content reconstruction in this study and the corresponding reconstruction from Wu et al. (2025) from 1880 to 2000 for 0–300 m and 300–2000 m.

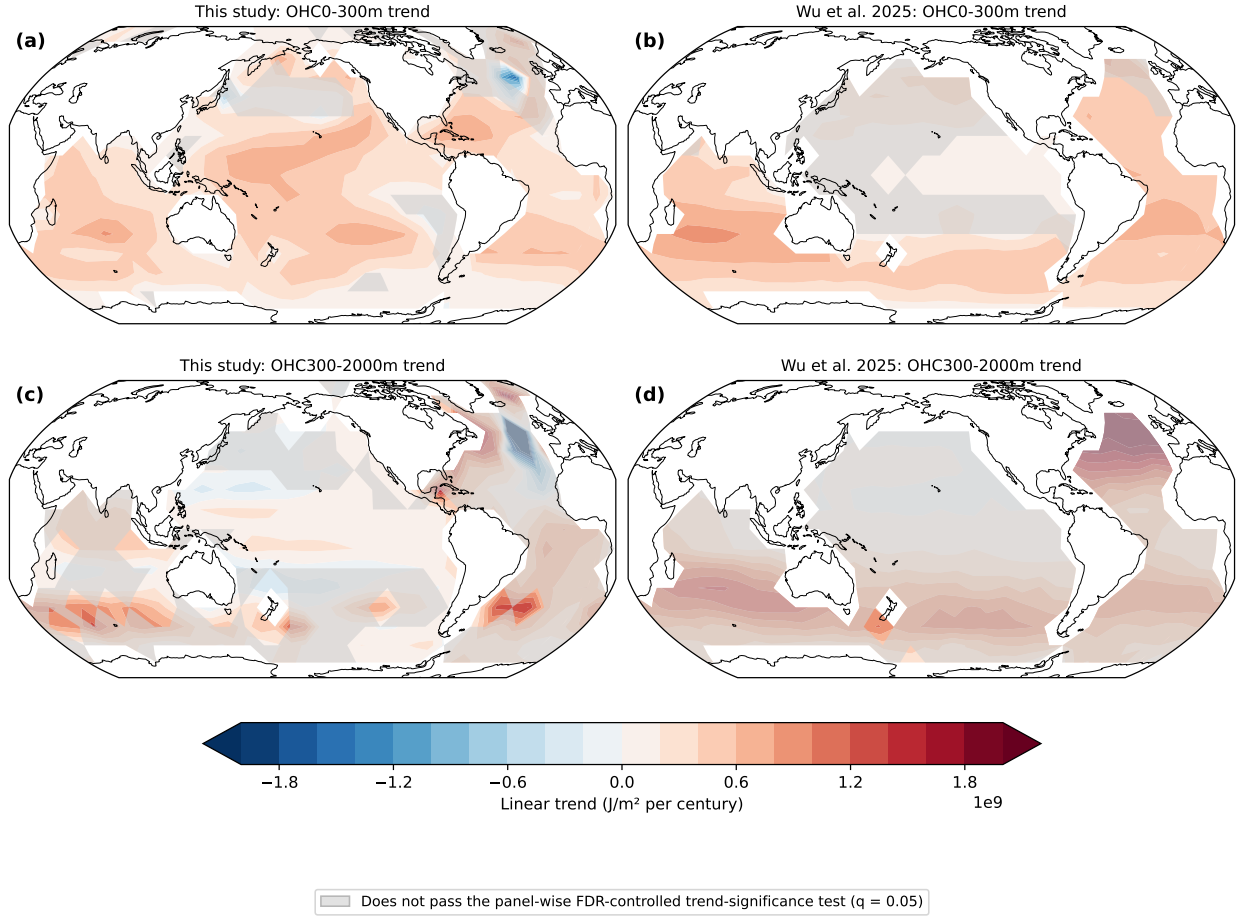


Figure S4: Comparison of 1880–2000 CE ocean heat content trends in this study and Wu et al. (2025). Panels compare trends from this study with the corresponding trends from Wu et al. (2025) for (a and b) OHC(0–300 m) and (c and d) OHC(300–2000 m). Semitransparent gray overlays identify grid cells that do not pass the panel-wise FDR-controlled trend-significance test at $q = 0.05$; unmasked colored regions pass the test described in Supplementary Section 8. The FDR-significant regions represent 73.4%, 59.2%, 23.1%, and 0.4% of the ocean area in (a) to (d), respectively. Only two isolated grid cells pass the test in (d), providing no evidence for a large-scale significant trend pattern in the Wu et al. (2025) 300–2000-m field.

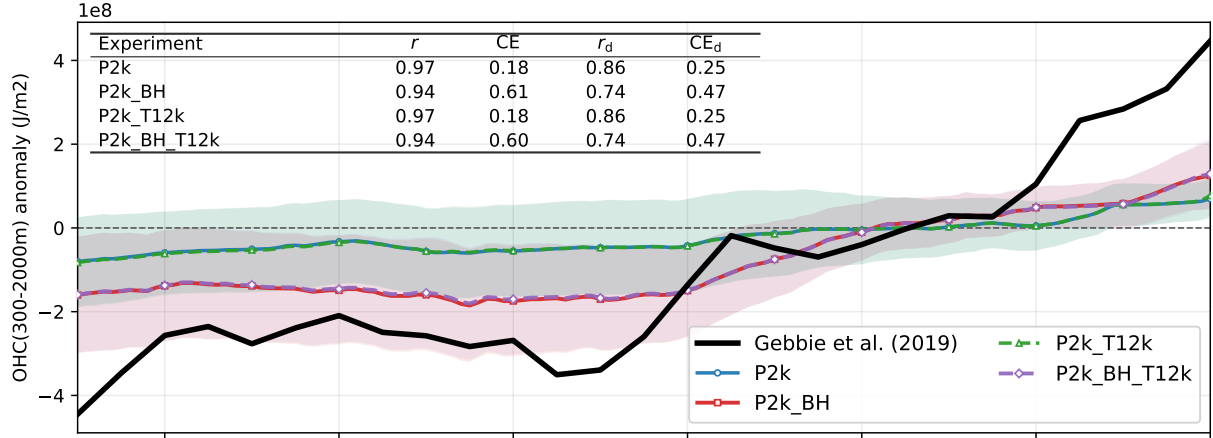
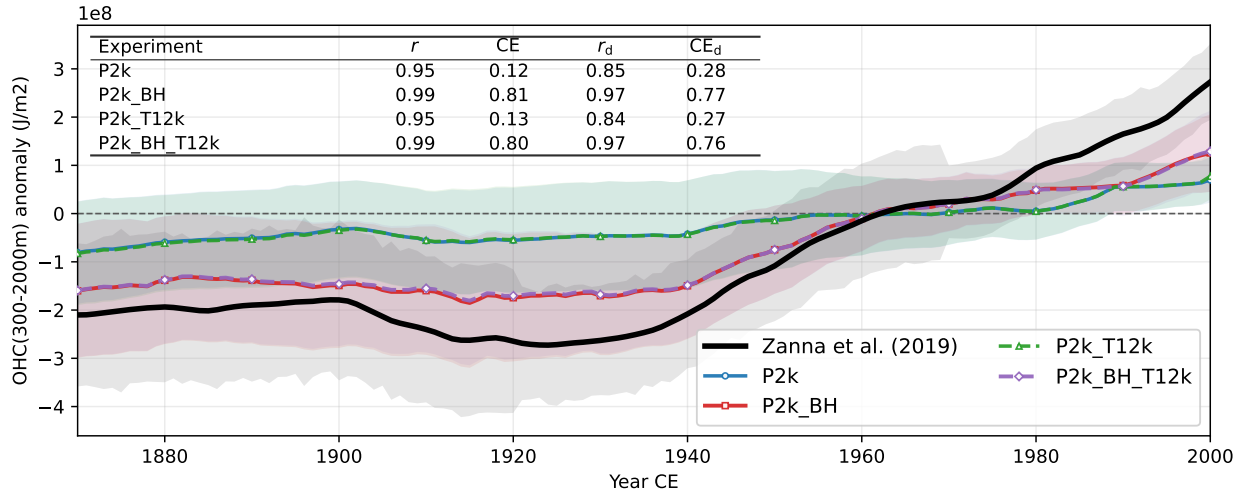
(a) Comparison of OHC(300–2000 m) reconstructions with Gebbie et al. (2019)**(b) Comparison of OHC(300–2000 m) reconstructions with Zanna et al. (2019)**

Figure S5: Comparison of reconstructed global-mean OHC(300–2000 m) anomalies with external estimates. (a) Comparison of the P2k, P2k_BH, P2k_T12k, and P2k_BH_T12k reconstructions with Gebbie and Huybers (2019). (b) Corresponding comparison with Zanna et al. (2019). Colored lines show the mean across the five model-emulator-specific reconstructions for each experiment, and colored shading denotes ± 1 standard deviation across these reconstructions. Black lines show the external estimates; gray shading in (b) denotes the reported uncertainty. In both panels, P2k closely overlaps P2k_T12k, while P2k_BH closely overlaps P2k_BH_T12k. All anomalies are referenced to the 1950–1980 CE climatology. Inset tables report Pearson correlation coefficients (r), coefficients of efficiency (CE), and their detrended counterparts (r_d and CE_d). Detrended metrics are calculated after linearly detrending each reconstruction-reference pair over their common interval.

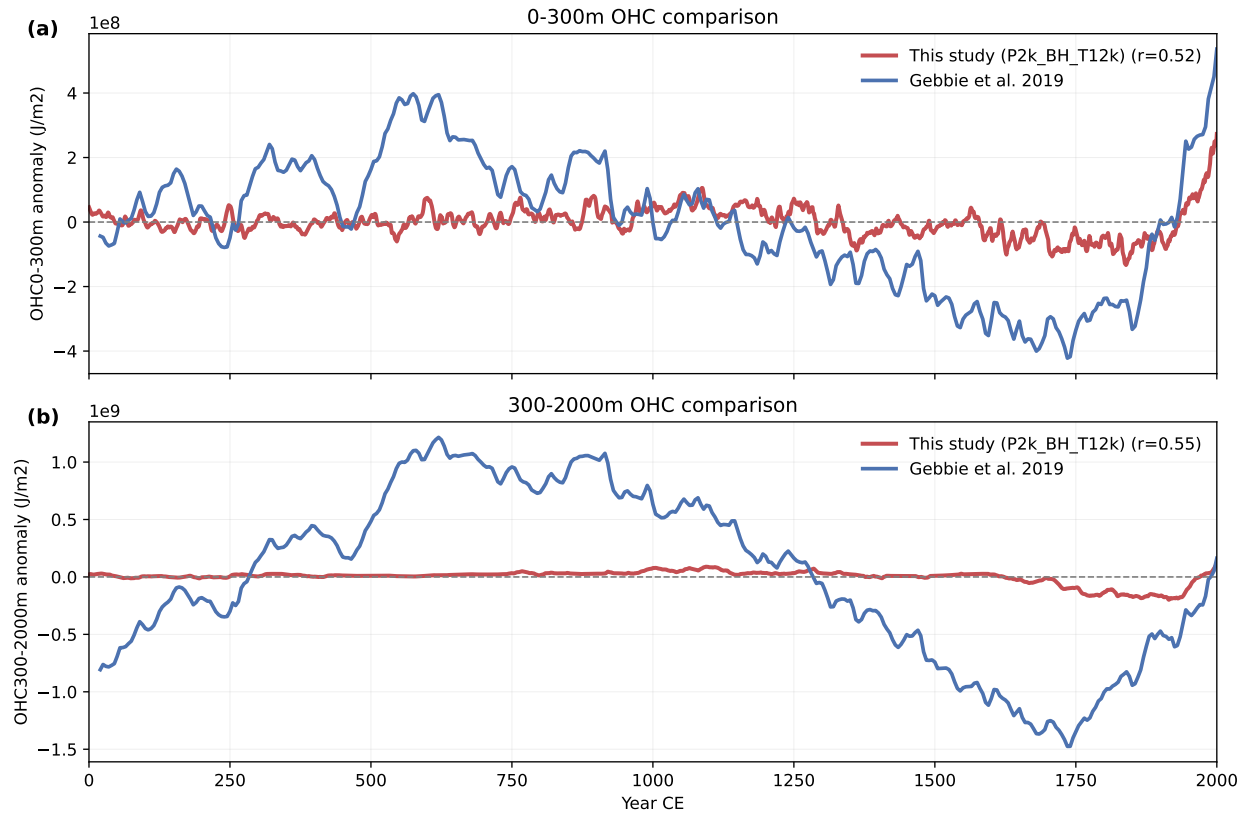


Figure S6: Comparison of reconstructed upper-300 m and 300–2000 m ocean heat content with an external reconstruction. The ocean heat content anomaly from the P2k_BH_T12k reconstruction is compared with the product derived from Gebbie and Huybers (2019).

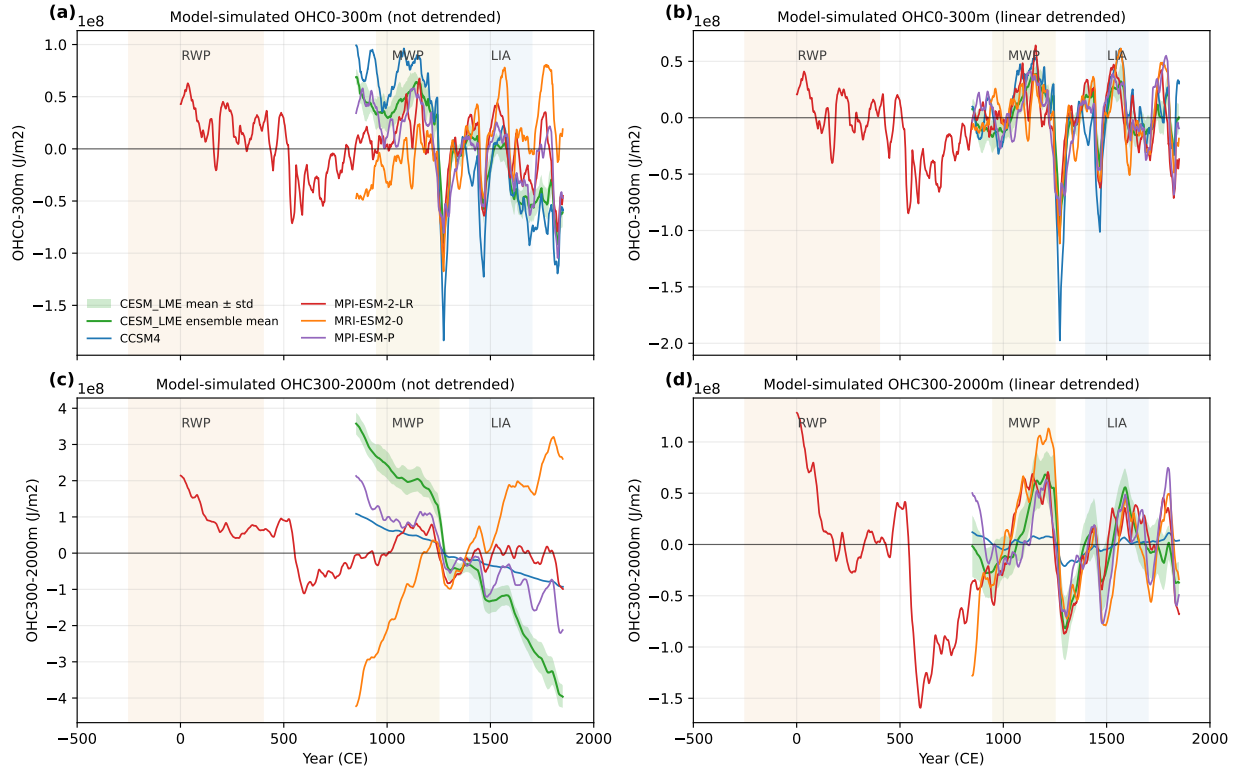


Figure S7: Layer-resolved model-simulated ocean heat-content changes. Upper panels show OHC(0–300 m), and lower panels show OHC(300–2000 m). Left panels show the original model-simulated global-mean OHC series, and right panels show the same series after linear detrending. All series are anomalies relative to the 850–1850 CE mean and are smoothed with a 30-year centered running mean. CESM-LME is shown as the ensemble mean, with shading indicating ± 1 standard deviation; other model simulations are shown as individual lines. Shaded vertical bands mark the Roman Warm Period (RWP), Medieval Warm Period (MWP), and Little Ice Age (LIA).

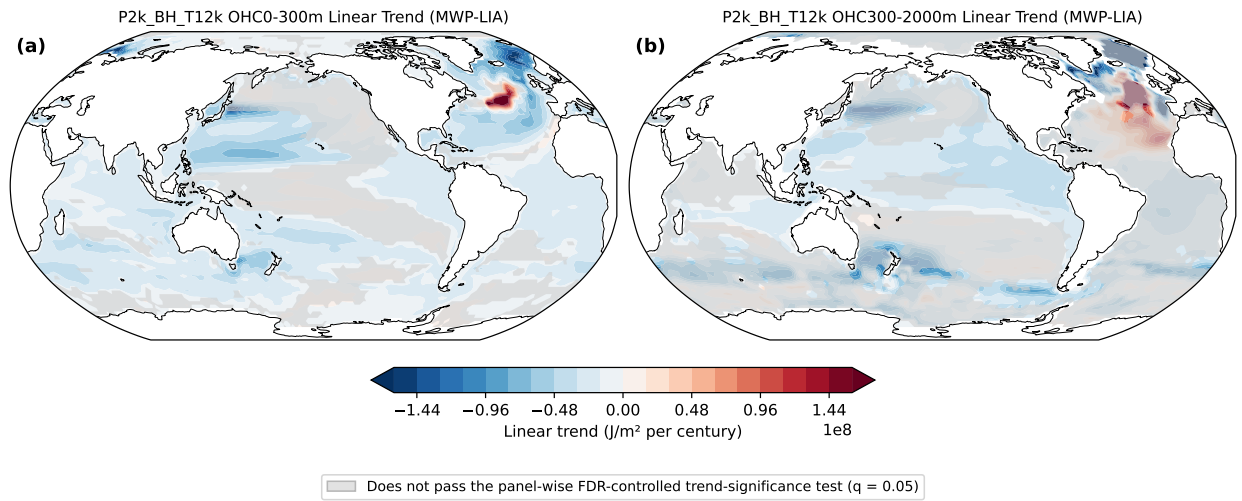


Figure S8: Linear trends in ocean heat content across the Medieval Warm Period to Little Ice Age transition. Spatial patterns show trends from the P2k_BH_T12k reconstruction for (a) OHC(0–300 m) and (b) OHC(300–2000 m) over the interval spanning the Medieval Warm Period to the Little Ice Age. Semitransparent gray overlays identify grid cells that do not pass the panel-wise FDR-controlled trend-significance test at $q = 0.05$; unmasked colored regions pass the test described in Supplementary Section 8. The FDR-significant regions represent 64.4% and 26.3% of the ocean area in (a) and (b), respectively.

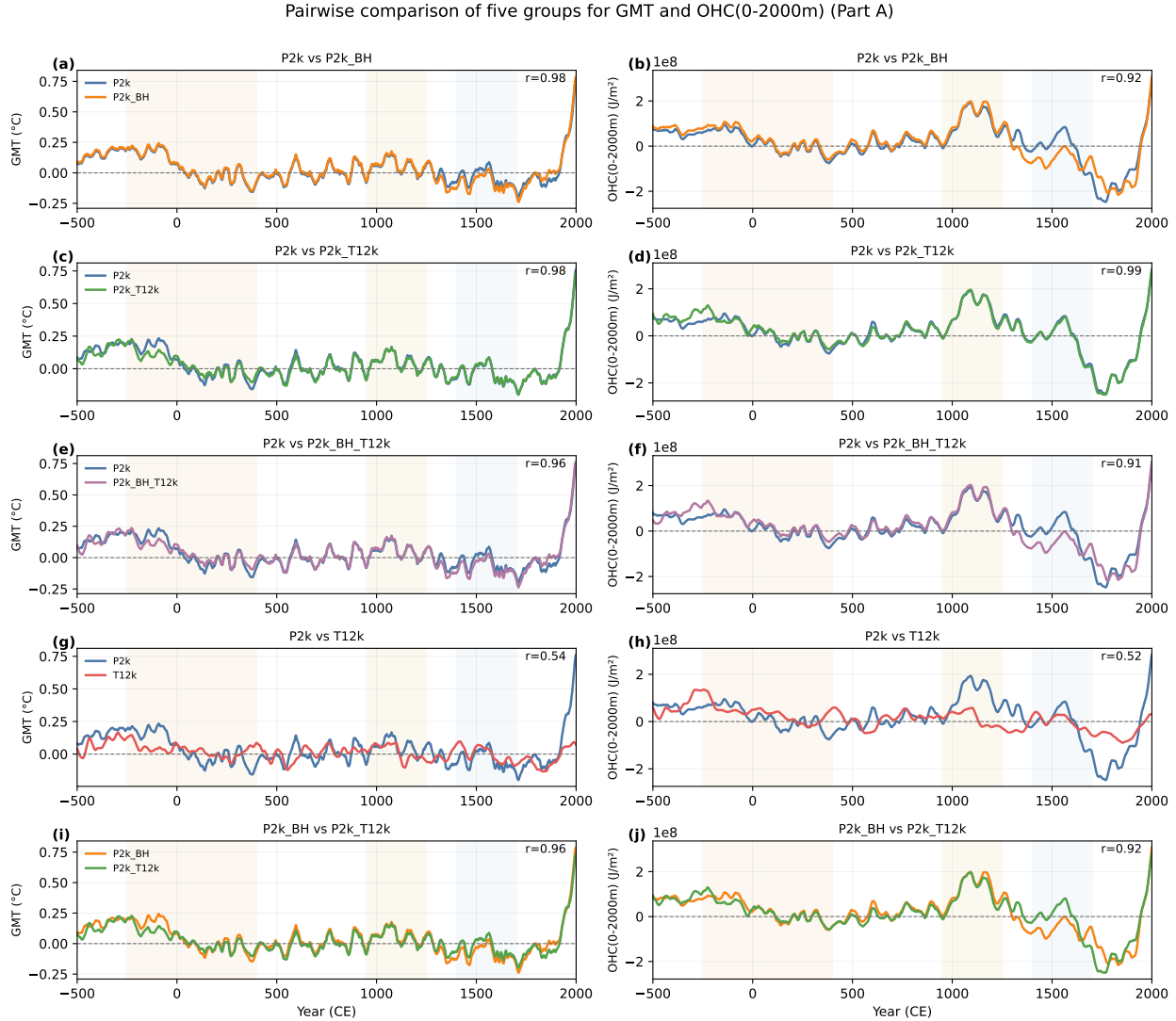


Figure S9: Pairwise comparisons of global mean temperature and upper-2000-m ocean heat content across reconstruction experiments, part A. Each row shows one pairwise comparison between two experiments. Left panels show 30-year low-pass-filtered global mean temperature (GMT) anomalies, and right panels show 30-year low-pass-filtered upper-2000-m ocean heat content anomalies, computed as $OHC_{0-300} + OHC_{300-2000}$. The five experiments are P2k = PAGES2k only, P2k_BH = PAGES2k + boreholes, P2k_T12k = PAGES2k + Temp12k, P2k_BH_T12k = PAGES2k + boreholes + Temp12k, and T12k = Temp12k only ($J_m = 1$). Correlation coefficients for each pair are shown in the panel corners. All anomalies are referenced to the 0–2000 CE climatology, and the Roman Warm Period (RWP), Medieval Warm Period (MWP), and Little Ice Age (LIA) are indicated.

Pairwise comparison of five groups for GMT and OHC(0-2000m) (Part B)

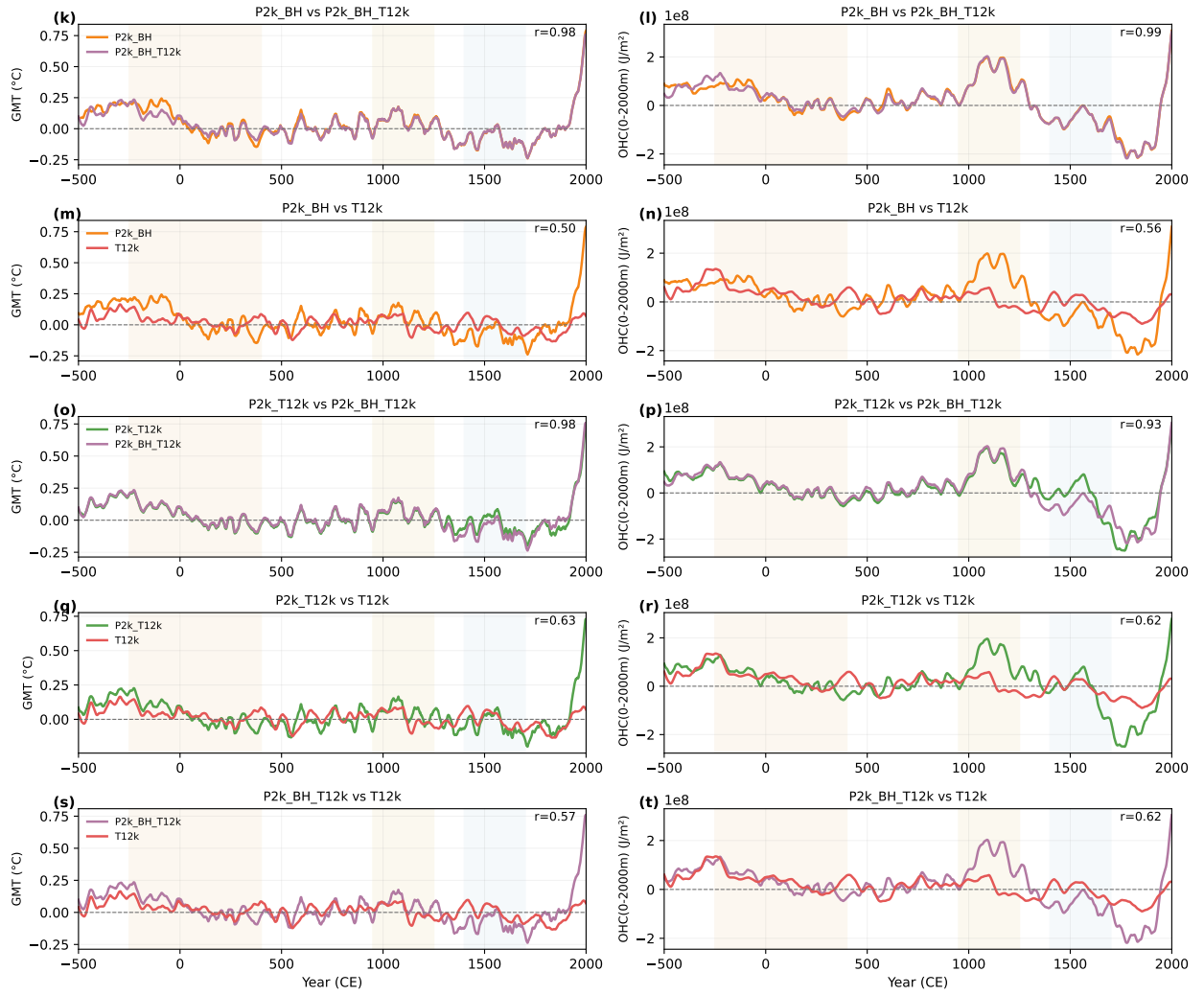


Figure S10: Pairwise comparisons of global mean temperature and upper-2000-m ocean heat content across reconstruction experiments, part B. Continued from Fig. S9.

References

- Amrhein, D. E., G. J. Hakim, and L. A. Parsons, 2020: Quantifying structural uncertainty in paleoclimate data assimilation with an application to the last millennium. *Geophysical Research Letters*, **47** (22), e2020GL090485.
- Benjamini, Y., and Y. Hochberg, 1995: Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, **57** (1), 289–300, <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>.
- Bradley, R. S., 1999: *Paleoclimatology: reconstructing climates of the Quaternary*, Vol. 68. Elsevier.
- Christiansen, B., and F. C. Ljungqvist, 2017: Challenges and Perspectives for Large-Scale Temperature Reconstructions of the Past Two Millennia. *Reviews of Geophysics*, **55** (1), 40–96.
- Council, N. R., D. on Earth, L. Studies, B. on Atmospheric Sciences, and . Y. Committee on Surface Temperature Reconstructions for the Last 2, 2007: *Surface temperature reconstructions for the last 2,000 years*. National Academies Press.
- Courtier, P., J.-N. Thépaut, and A. Hollingsworth, 1994: A strategy for operational implementation of 4d-var, using an incremental approach. *Quarterly Journal of the Royal Meteorological Society*, **120** (519), 1367–1387, <https://doi.org/10.1002/qj.49712051912>.
- Cuesta-Valero, F. J., A. García-García, H. Beltrami, J. F. González-Rouco, E. García-Bustamante, and J. P. Montávez, 2021: Long-term global ground heat flux and continental heat storage from geothermal data. *Climate of the Past*, **17** (1), 451–468, <https://doi.org/10.5194/cp-17-451-2021>.
- Dolman, A. M., and T. Laepple, 2018: Sedproxy: a forward model for sediment-archived climate proxies. *Climate of the Past*, **14** (12), 1851–1868.
- Emile-Geay, J., G. J. Hakim, F. Viens, F. Zhu, and D. E. Amrhein, 2025: Temporal comparisons involving paleoclimate data assimilation: Challenges and remedies. *Journal of Climate*, **38** (5), 1365–1385, <https://doi.org/10.1175/JCLI-D-24-0101.1>.
- Erb, M. P., N. P. McKay, N. Steiger, S. Dee, C. Hancock, R. F. Ivanovic, L. J. Gregoire, and P. Valdes, 2022: Reconstructing holocene temperatures in time and space using paleoclimate data assimilation. *Climate of the Past*, **18** (12), 2599–2629.
- Evans, M. N., S. E. Tolwinski-Ward, D. M. Thompson, and K. J. Anchukaitis, 2013: Applications of proxy system modeling in high resolution paleoclimatology. *Quaternary Science Reviews*, **76** (0), 16–28, <https://doi.org/10.1016/j.quascirev.2013.05.024>.
- Evensen, G., F. C. Vossepoel, and P. J. van Leeuwen, 2022: *Data Assimilation Fundamentals: A Unified Formulation of the State and Parameter Estimation Problem*. Springer Textbooks in Earth Sciences, Geography and Environment, Springer, Cham, <https://doi.org/10.1007/978-3-030-96709-3>.
- Fisher, M., M. Leutbecher, and G. A. Kelly, 2005: On the equivalence between kalman smoothing and weak-constraint four-dimensional variational data assimilation. *Quarterly Journal of the Royal Meteorological Society*, **131** (613), 3235–3246, <https://doi.org/10.1256/qj.04.142>.
- Fisher, M., Y. Trémolet, H. Auvinen, D. Tan, and P. Poli, 2011: Weak-constraint and long-window 4D-Var. ECMWF Technical Memorandum 655, European Centre for Medium-Range Weather Forecasts. <https://doi.org/10.21957/9ii4d4dsq>.
- Franke, J., S. Brönnimann, J. Bhend, and Y. Brugnara, 2017: A monthly global paleo-reanalysis of the atmosphere from 1600 to 2005 for studying past climatic variations. *Scientific data*, **4** (1), 1–19.
- Gebbie, G., and P. Huybers, 2019: The little ice age and 20th-century deep Pacific cooling. *Science*, **363** (6422), 70–74.
- Giorgetta, M. A., and Coauthors, 2013: Climate and carbon cycle changes from 1850 to 2100 in MPI-ESM simulations for the coupled model intercomparison project phase 5. *Journal of Advances in Modeling Earth Systems*, **5** (3), 572–597, <https://doi.org/10.1002/jame.20038>.
- Gleckler, P. J., B. D. Santer, C. M. Domingues, and Coauthors, 2012: Human-induced global ocean warming on multidecadal timescales. *Nature Climate Change*, **2**, 524–529, <https://doi.org/10.1038/nclimate1553>.
- Hakim, G. J., J. Emile-Geay, E. J. Steig, D. Noone, D. M. Anderson, R. Tardif, N. Steiger, and W. A. Perkins, 2016: The last millennium climate reanalysis project: Framework and first results. *Journal of Geophysical Research: Atmospheres*, **121** (12), 6745–6764.

- Holm, S., 1979: A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, **6** (2), 65–70, <https://doi.org/10.2307/4615733>.
- Huang, B., and Coauthors, 2017: Extended reconstructed sea surface temperature, version 5 (ERSSTv5): upgrades, validations, and intercomparisons. *Journal of Climate*, **30** (20), 8179–8205.
- Huang, S., H. N. Pollack, and P.-Y. Shen, 2000: Temperature trends over the past five centuries reconstructed from borehole temperatures. *Nature*, **403** (6771), 756–758.
- IPCC, 2021: *Climate Change 2021: The Physical Science Basis*. Cambridge University Press, <https://doi.org/10.1017/9781009157896>.
- IPCC, 2023: Climate change 2023: Synthesis report, summary for policymakers. Intergovernmental Panel on Climate Change.
- Ishii, M., Y. Fukuda, S. Hirahara, S. Yasui, T. Suzuki, and K. Sato, 2017: Accuracy of global upper ocean heat content estimation expected from present observational data sets. *Sola*, **13**, 163–167.
- Jones, P., and Coauthors, 2009: High-resolution palaeoclimatology of the last millennium: a review of current status and future prospects. *The Holocene*, **19** (1), 3–49, <https://doi.org/10.1177/0959683608098952>.
- Kaufman, D., and Coauthors, 2020: A global database of holocene paleotemperature records. *Scientific data*, **7** (1), 115.
- Landrum, L., B. L. Otto-Bliesner, E. R. Wahl, A. Conley, P. J. Lawrence, N. Rosenbloom, and H. Teng, 2013: Last millennium climate and its variability in CCSM4. *Journal of Climate*, **26** (4), 1085–1111, <https://doi.org/10.1175/JCLI-D-11-00326.1>.
- Le Dimet, F.-X., and O. Talagrand, 1986: Variational algorithms for analysis and assimilation of meteorological observations: Theoretical aspects. *Tellus A*, **38A** (2), 97–110, <https://doi.org/10.3402/tellusa.v38i2.11706>.
- Lenssen, N. J., G. A. Schmidt, J. E. Hansen, M. J. Menne, A. Persin, R. Ruedy, and D. Zyss, 2019: Improvements in the GISTEMP uncertainty model. *Journal of Geophysical Research: Atmospheres*, **124** (12), 6307–6326.
- Liu, D. C., and J. Nocedal, 1989: On the limited memory bfgs method for large scale optimization. *Mathematical programming*, **45** (1), 503–528.
- Lorenc, A. C., 2003: The potential of the ensemble kalman filter for nwp—a comparison with 4d-var. *Quarterly Journal of the Royal Meteorological Society*, **129** (595), 3183–3203, <https://doi.org/10.1256/qj.02.132>.
- Mann, M. E., and Coauthors, 2009: Global Signatures and Dynamical Origins of the Little Ice Age and Medieval Climate Anomaly. *Science*, **326** (5957), 1256–1260.
- Mauritsen, T., and Coauthors, 2019: Developments in the MPI-M earth system model version 1.2 (MPI-ESM1.2) and its response to increasing CO₂. *Journal of Advances in Modeling Earth Systems*, **11** (4), 998–1038, <https://doi.org/10.1029/2018MS001400>.
- Meng, Z., and G. J. Hakim, 2024: Reconstructing the tropical Pacific upper ocean using online data assimilation with a deep learning model. *Journal of Advances in Modeling Earth Systems*, **16** (11), e2024MS004422.
- Meng, Z., G. J. Hakim, and E. J. Steig, 2025: Coupled seasonal data assimilation of sea ice, ocean, and atmospheric dynamics over the last millennium. *Journal of Climate*, **38** (23), 7229–7247.
- Meng, Z., and T. Li, 2024: Why is the Pacific meridional mode most pronounced in boreal spring? *Climate Dynamics*, **62** (1), 459–471.
- Meng, Z., and L. M. Polvani, 2026: No evidence of statistically significant eurasian winter warming or cooling caused by large eruptions over the last millennium. *npj Climate and Atmospheric Science*, <https://doi.org/10.1038/s41612-026-01535-0>.
- Meng, Z., F. Zhu, and G. J. Hakim, 2023: Sacpy—A Python Package for Statistical Analysis of Climate. *AGU23*.
- Nash, J. E., and J. V. Sutcliffe, 1970: River flow forecasting through conceptual models part I—A discussion of principles. *Journal of hydrology*, **10** (3), 282–290.
- Ning, L., and Coauthors, 2026: Progress and prospects of paleoclimate data assimilation. *Science China Earth Sciences*, **69** (5), 1605–1617.
- Okazaki, A., T. Miyoshi, K. Yoshimura, S. J. Greybush, and F. Zhang, 2021: Revisiting online and offline data assimilation comparison for paleoclimate reconstruction: An idealized OSSE study. *Journal of Geophysical Research: Atmospheres*, **126** (16), e2020JD034214.

- O'Neill, B. C., and Coauthors, 2016: The scenario model intercomparison project (scenariomip) for cmip6. *Geoscientific Model Development*, **9** (9), 3461–3482.
- Orsi, A. J., B. D. Cornuelle, and J. P. Severinghaus, 2012: Little Ice Age cold interval in West Antarctica: evidence from borehole temperature at the West Antarctic Ice Sheet (WAIS) divide. *Geophysical Research Letters*, **39** (9).
- Osman, M. B., J. E. Tierney, J. Zhu, R. Tardif, G. J. Hakim, J. King, and C. J. Poulsen, 2021: Globally resolved surface temperatures since the last glacial maximum. *Nature*, **599** (7884), 239–244.
- Otto-Bliesner, B. L., and Coauthors, 2016: Climate variability and change since 850 CE: An ensemble approach with the Community Earth System Model. *Bulletin of the American Meteorological Society*, **97** (5), 735–754, <https://doi.org/10.1175/BAMS-D-14-00233.1>.
- PAGES2k Consortium and others, 2017: A global multiproxy database for temperature reconstructions of the Common Era. *Scientific data*, **4**.
- Parsons, L. A., D. E. Amrhein, S. C. Sanchez, R. Tardif, M. K. Brennan, and G. J. Hakim, 2021: Do multi-model ensembles improve reconstruction skill in paleoclimate data assimilation? *Earth and Space Science*, **8** (4), e2020EA001 467.
- Paszke, A., and Coauthors, 2019: Pytorch: An imperative style, high-performance deep learning library. *Advances in neural information processing systems*, **32**.
- Penland, C., and T. Magorian, 1993a: Prediction of Niño 3 sea surface temperatures using linear inverse modeling. *Journal of Climate*, **6** (6), 1067–1076.
- Penland, C., and T. Magorian, 1993b: Prediction of Niño 3 Sea Surface Temperatures Using Linear Inverse Modeling. *Journal of Climate*, **6** (6), 1067–1076, [https://doi.org/10.1175/1520-0442\(1993\)006<1067:PONSST>2.0.CO;2](https://doi.org/10.1175/1520-0442(1993)006<1067:PONSST>2.0.CO;2).
- Penland, C., and L. Matrosova, 1994: A balance condition for stochastic numerical models with application to the El Niño–Southern Oscillation. *Journal of climate*, **7** (9), 1352–1372.
- Perkins, W., and G. Hakim, 2021: Coupled atmosphere–ocean reconstruction of the last millennium using online data assimilation. *Paleoceanography and Paleoclimatology*, **36** (5), e2020PA003 959.
- Perkins, W. A., and G. Hakim, 2020: Linear Inverse Modeling for Coupled Atmosphere–Ocean Ensemble Climate Prediction. *Journal of Advances in Modeling Earth Systems*, **12** (1), e2019MS001 778.
- Perkins, W. A., and G. J. Hakim, 2017: Reconstructing paleoclimate fields using online data assimilation with a linear inverse model. *Climate of the Past*, **13** (5), 421–436.
- Pollack, H. N., and S. Huang, 2000: Climate reconstruction from subsurface temperatures. *Annual Review of Earth and Planetary Sciences*, **28** (1), 339–365, <https://doi.org/10.1146/annurev.earth.28.1.339>.
- Rabier, F., H. Järvinen, E. Klinker, J.-F. Mahfouf, and A. Simmons, 2000: The ecmwf operational implementation of four-dimensional variational assimilation. i: Experimental results with simplified physics. *Quarterly Journal of the Royal Meteorological Society*, **126** (564), 1143–1170, <https://doi.org/10.1256/smsqj.56414>.
- Rohde, R. A., and Z. Hausfather, 2020: The Berkeley Earth land/ocean temperature record. *Earth System Science Data*, **12** (4), 3469–3479.
- Santer, B. D., T. M. L. Wigley, J. S. Boyle, D. J. Gaffen, J. J. Hnilo, D. Nychka, D. E. Parker, and K. E. Taylor, 2000: Statistical significance of trends and trend differences in layer-average atmospheric temperature time series. *Journal of Geophysical Research: Atmospheres*, **105** (D6), 7337–7356, <https://doi.org/10.1029/1999JD901105>.
- Sasaki, Y., 1970: Some basic formalisms in numerical variational analysis. *Monthly Weather Review*, **98** (12), 875–883, [https://doi.org/10.1175/1520-0493\(1970\)098<0875:SBFINV>2.3.CO;2](https://doi.org/10.1175/1520-0493(1970)098<0875:SBFINV>2.3.CO;2).
- Steig, E. J., and A. J. Orsi, 2013: The heat is on in antarctica. *Nature Geoscience*, **6** (2), 87–88, <https://doi.org/10.1038/ngeo1717>.
- Steiger, N. J., J. E. Smerdon, E. R. Cook, and B. I. Cook, 2018: A reconstruction of global hydroclimate and dynamical variables over the Common Era. *Scientific data*, **5** (1), 1–15.
- Talagrand, O., and P. Courtier, 1987: Variational assimilation of meteorological observations with the adjoint vorticity equation. i: Theory. *Quarterly Journal of the Royal Meteorological Society*, **113** (478), 1311–1328, <https://doi.org/10.1002/qj.49711347812>.
- Tardif, R., and Coauthors, 2019: Last Millennium Reanalysis with an expanded proxy database and seasonal proxy modeling. *Climate of the Past*, **15** (4), 1251–1273.
- Tierney, J. E., E. J. Judd, M. B. Osman, J. M. King, O. J. Truax, N. J. Steiger, D. E. Amrhein, and K. J. Anchukaitis, 2025: Advances in Paleoclimate Data Assimilation. *Annual Review of Earth and Planetary Sciences*, **53**.

- Tierney, J. E., J. Zhu, J. King, S. B. Malevich, G. J. Hakim, and C. J. Poulsen, 2020: Glacial cooling and climate sensitivity revisited. *Nature*, **584** (7822), 569–573, <https://doi.org/10.1038/s41586-020-2617-x>.
- Toohey, M., and M. Sigl, 2017: Volcanic stratospheric sulfur injections and aerosol optical depth from 500 BCE to 1900 CE. *Earth System Science Data*, **9** (2), 809–831.
- Trémolet, Y., 2006: Accounting for an imperfect model in 4d-var. *Quarterly Journal of the Royal Meteorological Society*, **132** (621), 2483–2504, <https://doi.org/10.1256/qj.05.224>.
- Valler, V., J. Franke, Y. Brugnara, and S. Brönnimann, 2022: An Updated Global Atmospheric Paleo-reanalysis Covering the Last 400 Years. *Geoscience Data Journal*, **9** (1), 89–107, <https://doi.org/10.1002/gdj3.121>.
- Valler, V., and Coauthors, 2024: ModE-RA: a global monthly paleo-reanalysis of the modern era 1421 to 2008. *Scientific data*, **11** (1), 36.
- Widmann, M., H. Goosse, G. van der Schrier, R. Schnur, and J. Barkmeijer, 2010: Using data assimilation to study extratropical Northern Hemisphere climate over the last millennium. *Climate of the Past*, **6** (5), 627–644.
- Wilks, D. S., 2011: *Statistical Methods in the Atmospheric Sciences*. 3rd ed., Academic Press, Oxford.
- Wilks, D. S., 2016: “The stippling shows statistically significant grid points”: How research results are routinely overstated and overinterpreted, and what to do about it. *Bulletin of the American Meteorological Society*, **97** (12), 2263–2273, <https://doi.org/10.1175/BAMS-D-15-00267.1>.
- Wu, Q., J. M. Gregory, L. Zanna, and S. Khatiwala, 2025: Time-varying global energy budget since 1880 from a reconstruction of ocean warming. *Proceedings of the National Academy of Sciences of the United States of America*, **122** (20), e2408839 122, <https://doi.org/10.1073/pnas.2408839122>.
- Yukimoto, S., and Coauthors, 2019: The meteorological research institute earth system model version 2.0, MRI-ESM2.0: Description and basic evaluation of the physical component. *Journal of the Meteorological Society of Japan. Ser. II*, **97** (5), 931–965, <https://doi.org/10.2151/jmsj.2019-051>.
- Zanna, L., S. Khatiwala, J. M. Gregory, J. Ison, and P. Heimbach, 2019: Global reconstruction of historical ocean heat storage and transport. *Proceedings of the National Academy of Sciences*, **116** (4), 1126–1131, <https://doi.org/10.1073/pnas.1808838115>.
- Zhang, H., M. Li, and Y. Hu, 2025: Paleoclimate Data Assimilation: Principles and Prospects. *Science China Earth Sciences*, 1–18.
- Zhu, F., and Coauthors, 2019: Climate models can correctly simulate the continuum of global-average temperature variability. *Proceedings of the National Academy of Sciences of the United States of America*, **116** (18), 8728–8733, <https://doi.org/10.1073/pnas.1809959116>.